

ME23312 ENGINEERING THERMODYNAMICS

UNIT III – Properties of Pure Substance • Lecture Notes

UNIT 3 • LECTURE NOTES

Unit III – Properties of Pure Substance

Detailed lecture notes — formation of steam, property diagrams and the p - v - T surface, steam tables and the Mollier chart, dryness fraction and calorimeters, work and heat transfer in non-flow and flow processes of steam, first- and second-law applications, with solved numericals

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SYLLABUS

UNIT III — PROPERTIES OF PURE SUBSTANCE (9 + 3 periods)

Steam – formation and its thermodynamic properties – p - v , p - T , T - v , T - s , h - s diagrams. PVT surface. Determination of dryness fraction. Calculation of work done and heat transfer in non-flow and flow processes using Steam Table and Mollier Chart. Application of I and II law for pure substances.

Course Outcome CO3: Apply thermodynamic principles to analyze the properties of pure substance, utilizing diagrams and charts to calculate work, heat transfer, and dryness fraction.

Text: P. K. Nag, *Engineering Thermodynamics*, 6th Edn., McGraw-Hill (Chapter 9). **References:** Cengel & Boles, *Thermodynamics – An Engineering Approach*, 8th Edn.; R. K. Rajput, *A Text Book of Engineering Thermodynamics*, 5th Edn.; steam tables (IAPWS-97 values are used throughout these notes).

CONTENTS

1. Introduction – pure substances and their phases
2. Formation of steam at constant pressure – the T - v and T - Q diagrams
3. Property diagrams of a pure substance – p - v - T surface, p - T , p - v , T - s and h - s (Mollier) diagrams
4. Steam tables and the properties of wet, dry saturated and superheated steam (with worked examples)
5. Determination of the dryness fraction – calorimeters (with worked examples)
6. Non-flow processes of steam – work and heat transfer (with worked examples)
7. Flow processes of steam – turbine, nozzle, throttling and mixing (with worked examples)
8. Application of the first and second laws to pure substances
9. Practice problems
10. Summary and formula sheet
11. Review questions (two-mark and thirteen-mark)

Learning objectives. After studying this unit you should be able to: (i) define a pure substance and describe the changes of phase it undergoes on heating at constant pressure, using the $T-v$ and $T-Q$ diagrams; (ii) sketch and interpret the $p-v-T$ surface and its $p-T$, $p-v$, $T-v$, $T-s$ and $h-s$ projections, locating the triple point, the critical point and the saturation lines; (iii) define dryness fraction, wet, dry saturated and superheated steam, degree of superheat and degree of subcooling, sensible and latent heat; (iv) use the saturated and superheated steam tables, with interpolation, to find v , u , h and s of water and steam in any state; (v) explain the working of the separating, throttling and combined calorimeters and calculate the dryness fraction from their readings; (vi) calculate work and heat transfer for constant-volume, constant-pressure, isothermal, hyperbolic, isentropic and polytropic processes of steam in closed systems; (vii) apply the steady flow energy equation to steam turbines, nozzles, throttling valves and mixing chambers using the tables and the Mollier chart; (viii) apply the second law to steam processes – isentropic efficiency, entropy generation and the feasibility of a process.

1. Introduction – pure substances and their phases

1.1 Pure substance

A **pure substance** is one that has a homogeneous and invariable chemical composition throughout its mass, even though it may exist in more than one phase. Water is the pure substance of this unit: ice, liquid water and steam are all H_2O , and a mixture of water and steam in a boiler drum is still a pure substance because every part of it has the same chemical composition. Nitrogen, oxygen, carbon dioxide, ammonia and the refrigerants are pure substances. A mixture of gases such as air is treated as a pure substance so long as its composition does not change – but if air is cooled until some of it liquefies, the liquid is richer in oxygen than the vapour, the composition of the phases differs and the mixture is no longer a pure substance. A mixture of oil and water is not a pure substance because it is not chemically homogeneous.

1.2 Phases of a pure substance

Matter exists in three principal phases – **solid**, **liquid** and **gas (vapour)** – distinguished by the strength of the intermolecular forces. In a solid the molecules are held in a fixed lattice; in a liquid they move freely but remain close together, so the liquid has a definite volume but no definite shape; in a gas they are far apart and move at random, filling any container. A **vapour** is a gas that is near its condensation conditions – steam is water vapour. The change from one phase to another is a **phase change**, and the phase changes of interest in engineering are **melting** (solid $\rightarrow$ liquid, also called fusion), **freezing** (liquid $\rightarrow$ solid), **vaporisation** or **boiling** (liquid $\rightarrow$ vapour), **condensation** (vapour $\rightarrow$ liquid), and **sublimation** (solid $\rightarrow$ vapour directly, without passing through the liquid phase, which occurs at pressures below the triple-point pressure – dry ice, solid CO_2 , sublimates at atmospheric pressure, and so does ice in a vacuum or on a very cold dry day).

1.3 Why steam?

Water is cheap, plentiful, chemically stable and non-toxic, and it has a large latent heat, so it is the working fluid of almost all thermal power stations, of process heating in industry, and of many refrigeration and heating systems. Because steam near its saturation conditions does not obey the ideal-gas equation, its properties cannot be computed from $p v = RT$; they have been measured and are tabulated in **steam tables** and plotted on charts, of which the **Mollier ($h-s$) chart** is the most used. This unit teaches how steam is formed, how its properties are represented, and how the tables and charts are used to calculate work and heat transfer in closed and open systems.

2. Formation of steam at constant pressure – the T–v and T–Q diagrams

2.1 Heating water at constant pressure

Consider 1 kg of water at 30 °C in a cylinder fitted with a frictionless piston loaded so that the pressure is a constant 0.1 MPa (Figure 3.1, left, states a to e).

- **State a – compressed (subcooled) liquid.** The water is below its boiling point for the given pressure; it is a *compressed liquid* (its pressure is higher than the saturation pressure at 30 °C, 4.25 kPa) or, equivalently, a *subcooled liquid* (its temperature is lower than the saturation temperature at 0.1 MPa, 99.6 °C). As heat is added its temperature rises and its volume increases very slightly.
- **State b – saturated liquid.** At 99.6 °C the water is about to boil: any further addition of heat begins vaporisation. A liquid at the temperature at which it is about to vaporise is a **saturated liquid**. Its properties are denoted by the subscript f: v_f , u_f , h_f , s_f .
- **States b to d – wet steam (liquid–vapour mixture).** Further heating converts liquid into vapour at constant temperature and pressure; the volume increases enormously (from 0.001043 to 1.694 m³/kg at 0.1 MPa). Any intermediate state c is a two-phase mixture of saturated liquid and saturated vapour in equilibrium, called **wet steam**.
- **State d – dry saturated vapour.** When the last drop of liquid has vaporised the substance is **dry saturated steam** (subscript g: v_g , u_g , h_g , s_g) – vapour at the saturation temperature containing no moisture, on the point of condensing if any heat is removed.
- **State e – superheated vapour.** Further heating raises the temperature above the saturation temperature; the steam is **superheated**, and its temperature and volume both increase with heat addition.

The temperature at which a pure substance changes phase at a given pressure is the **saturation temperature** T_{sat} , and the pressure at which it changes phase at a given temperature is the **saturation pressure** p_{sat} . The two are uniquely related along the vapour-pressure curve: water boils at 99.6 °C at 0.1 MPa, at 179.9 °C at 1 MPa, at 311.0 °C at 10 MPa and at 374.1 °C at the critical pressure of 22.06 MPa. The saturation temperature increases with pressure, which is why a pressure cooker cooks faster and why boiler pressures are raised to obtain higher cycle temperatures. During a phase change T and p are not independent – fixing one fixes the other – so a single one of them does not fix the state of wet steam; a second property such as v, h, s or the dryness fraction is needed.

2.2 Sensible heat, latent heat and superheat

On the **T–Q diagram** (Figure 3.1, right) the heat supplied to 1 kg of ice at –10 °C at atmospheric pressure until it becomes steam at 150 °C is plotted against temperature.

- **A–B, sensible heat of ice:** the temperature rises from –10 °C to 0 °C; $Q = c_{\text{ice}}\Delta T = 2.093 \times 10 = 20.9$ kJ/kg.
- **B–C, latent heat of fusion:** ice melts at 0 °C with no change of temperature; $Q = h_{\text{sf}} = 334.5$ kJ/kg.
- **C–D, sensible heat of water:** the water is heated from 0 °C to 100 °C; $Q = c_w\Delta T = 4.187 \times 100 = 418.7$ kJ/kg (the enthalpy of saturated liquid h_f at 100 °C, 419.1 kJ/kg, differs slightly because c_w is not exactly constant).
- **D–E, latent heat of vaporisation:** the water boils at 100 °C; $Q = h_{\text{fg}} = 2257$ kJ/kg – the longest segment, more than five times the sensible heat of the water.
- **E–F, superheat:** the steam is heated from 100 °C to 150 °C; $Q = c_{\text{ps}}\Delta T \approx 2.0 \times 50 = 100$ kJ/kg.

The total heat required is about 20.9 + 334.5 + 418.7 + 2257 + 100 = 3131 kJ/kg. **Sensible heat** is heat that produces a change of temperature without a change of phase; **latent heat** is heat absorbed or released at constant temperature during a change of phase. The **latent heat of fusion** h_{sf} is the heat needed to melt unit mass of solid at constant temperature and pressure; the **latent heat (enthalpy) of**

vaporisation $h_{fg} = h_g - h_f$ is the heat needed to convert unit mass of saturated liquid into dry saturated vapour at constant pressure. Both depend on the pressure: h_{fg} is 2257.5 kJ/kg at 0.1 MPa, 2014.4 kJ/kg at 1 MPa, 1317.6 kJ/kg at 10 MPa and zero at the critical point (Worked Example 3.1).

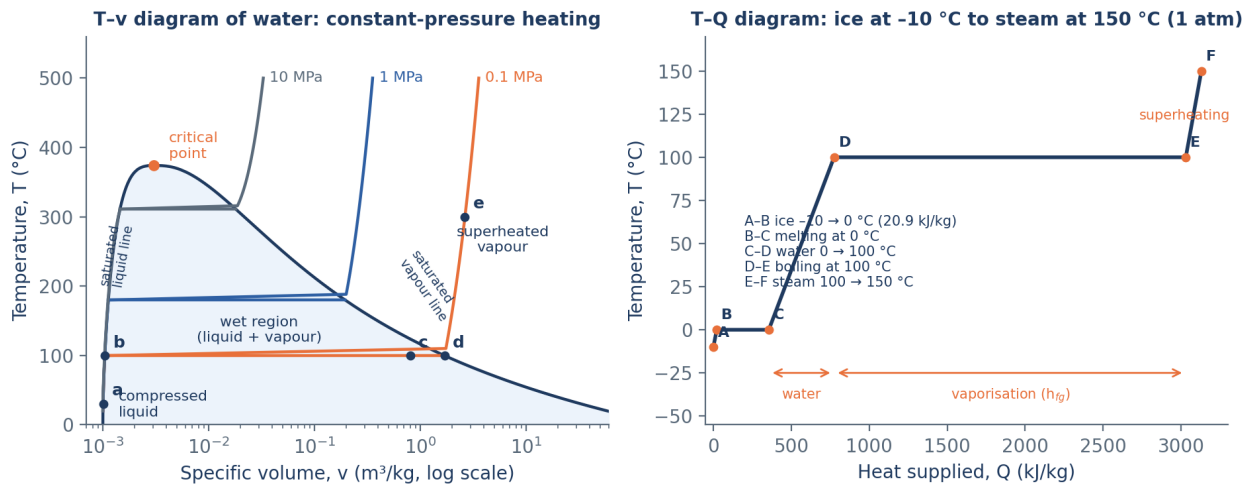


Figure 3.1 (Left) Formation of steam at constant pressure on the T - v diagram, with isobars at 0.1, 1 and 10 MPa and the saturation dome. (Right) Temperature-heat diagram for the conversion of ice at $-10\text{ }^{\circ}\text{C}$ into superheated steam at $150\text{ }^{\circ}\text{C}$ at atmospheric pressure.

2.3 Degree of superheat and degree of subcooling

The **degree of superheat** is the difference between the temperature of superheated steam and the saturation temperature at the same pressure, $T - T_{\text{sat}}$: steam at 1.2 MPa and $350\text{ }^{\circ}\text{C}$ has a degree of superheat of $350 - 188.0 = 162.0\text{ K}$. The **degree of subcooling** (or undercooling) of a compressed liquid is the difference between the saturation temperature at the given pressure and the actual temperature, $T_{\text{sat}} - T$: water at 1.2 MPa and $40\text{ }^{\circ}\text{C}$ is subcooled by $188.0 - 40 = 148.0\text{ K}$. A saturated liquid has zero subcooling and dry saturated steam zero superheat.

2.4 Evaporation and boiling

Evaporation is the change of liquid into vapour that takes place *at the free surface* at any temperature below the boiling point, because the more energetic surface molecules escape into the space above; it is slow, occurs without bubble formation, and its rate depends on the exposed area, the temperature and the vapour already present in the space (water in an open dish evaporates at room temperature). **Boiling** is vaporisation *throughout the bulk* of the liquid, with the formation of bubbles, which occurs only when the liquid has reached the saturation temperature corresponding to the pressure acting on it. The steam generated in a boiler is produced by boiling; the moisture picked up by air in a cooling tower is produced by evaporation.

2.5 The critical point and the triple point

As the pressure is raised the horizontal vaporisation segment b-d on the T - v diagram becomes shorter: the saturated liquid expands and the saturated vapour is compressed, so v_f and v_g approach each other, and so do h_f and h_g . At the **critical point** the saturated liquid and saturated vapour states coincide: the latent heat is zero, there is no distinction between liquid and vapour, and a meniscus is no longer visible. For water the critical point is $p_c = 22.064\text{ MPa}$, $T_c = 373.95\text{ }^{\circ}\text{C}$ (647.1 K), $v_c \approx 0.0031\text{ m}^3/\text{kg}$. Above the critical pressure a substance passes from a liquid-like to a gas-like state with no phase change and no boiling – *supercritical* boilers work in this region.

The **triple point** is the state at which the solid, liquid and vapour phases coexist in equilibrium; it is a single point on the p - T diagram and a line (the *triple line*) on the p - v and p - v - T diagrams. For water it is $T_{\text{tp}} = 0.01\text{ }^{\circ}\text{C}$ (273.16 K exactly) and $p_{\text{tp}} = 0.6117\text{ kPa}$; it is the fixed point that defines the Kelvin scale. Below the triple-point pressure a solid sublimates directly to vapour.

3. Property diagrams of a pure substance

The states of a pure substance can be represented on a three-dimensional p - v - T surface or, more conveniently, on its two-dimensional projections.

3.1 The p - v - T surface

Every equilibrium state of a pure substance is a point on a surface in p - v - T space, because any two of the three properties fix the third (Figure 3.2). The surface has three single-phase regions – **solid**, **liquid** and **vapour** – in which p , v and T can all vary independently, and three **two-phase regions** – solid–liquid, liquid–vapour and solid–vapour – in which the pressure and temperature are related (fixing one fixes the other) and only the volume changes as the proportion of the phases changes. The two-phase regions are *ruled surfaces*, made up of straight lines of constant p and T , and they appear on edge (as lines) when the surface is viewed along the v -axis. The three two-phase regions meet along the **triple line**; the liquid–vapour region narrows with increasing pressure and terminates at the **critical point**, above which the liquid and vapour regions merge.

Substances differ in the solid–liquid region. Most substances **contract on freezing** (the solid is denser than the liquid), so the solid–liquid surface lies at smaller volumes than the liquid and its projection on the p - T plane – the fusion curve – slopes to the right (melting point rises with pressure). Water is the important exception: it **expands on freezing** (ice floats), so the solid has the larger volume, the solid–liquid region leans the other way, and the fusion curve has a negative slope – the melting point of ice falls as the pressure rises, which is why ice under a skate blade melts.

p - v - T surface: substance that contracts on freezing

p - v - T surface: substance that expands on freezing (water)

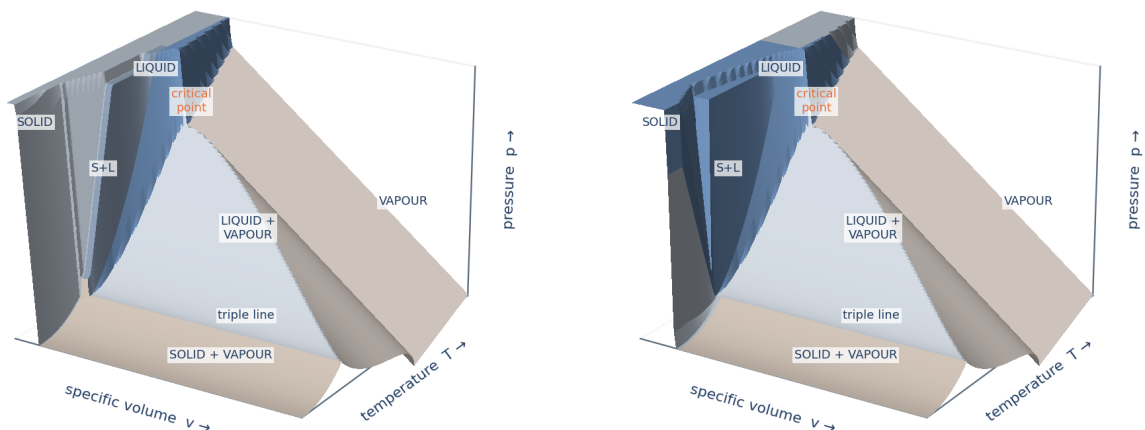


Figure 3.2 Schematic p - v - T surfaces of a substance that contracts on freezing (left) and of water, which expands on freezing (right), showing the single-phase regions, the two-phase regions, the triple line and the critical point.

3.2 The p - T (phase) diagram

Projecting the surface on the p - T plane makes each two-phase region collapse into a single curve (Figure 3.3, right): the **vaporisation (vapour-pressure) curve** separating liquid from vapour and ending at the critical point; the **fusion curve** separating solid from liquid (negative slope for water); and the **sublimation curve** separating solid from vapour. The three curves meet at the **triple point**. The diagram is called the phase diagram because it shows at a glance which phase exists at any p and T , and it is the diagram on which the Clausius–Clapeyron equation of Unit IV is written.

3.3 The p - v and T - v diagrams

On the p - v plane (Figure 3.3, left) the liquid–vapour region is a dome bounded by the **saturated liquid line** and the **saturated vapour line**, which meet at the critical point; the triple line is the horizontal base of the dome. Isotherms cross the dome horizontally (constant pressure during vaporisation), fall steeply

in the liquid region (a liquid is almost incompressible) and follow a hyperbola-like curve in the vapour region, approaching $p v = RT$ at low pressures. The **critical isotherm** has a horizontal point of inflexion at the critical point. The T – v diagram (Figure 3.1, left) is the same picture with constant-pressure lines instead of isotherms.

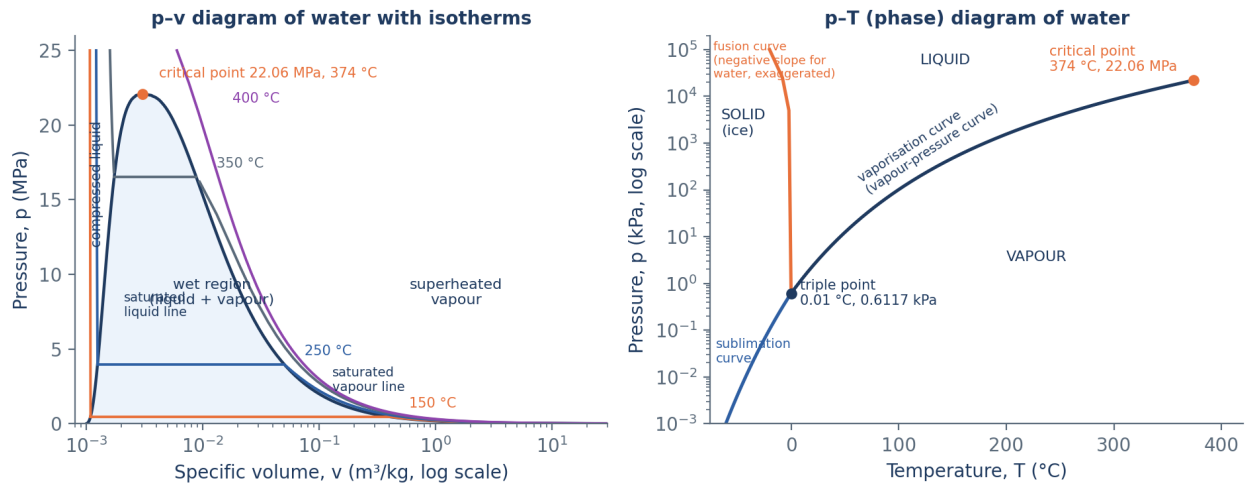


Figure 3.3 (Left) The p – v diagram of water with the saturation dome and isotherms. (Right) The p – T phase diagram of water with the triple point and the critical point.

3.4 The T – s diagram

The T – s diagram (Figure 3.4, left) is the working diagram for second-law analysis of steam plants, because the area under a reversible process is the heat transferred and an isentropic process is a vertical line. The saturation dome has the same form as on the p – v plane; constant-pressure lines coincide with the isotherms inside the dome, rise steeply in the liquid region (almost along the saturated liquid line) and curve upwards in the superheated region. Lines of constant dryness fraction divide the horizontal segments of the dome in proportion. The entropy of water is measured from a datum of $s = 0$ for saturated liquid at the triple point.

3.5 The h – s (Mollier) diagram

The **Mollier chart** (Figure 3.4, right) plots specific enthalpy against specific entropy. It is the chart of choice for flow calculations because the quantities needed – the enthalpy drop in a turbine or nozzle (a vertical distance for an isentropic process) and the end state after throttling (a horizontal line, $h = \text{constant}$) – can be read directly. The families of lines drawn on it are: constant-pressure lines (straight in the wet region, since $dh/ds = T$ at constant p , and curving upwards in the superheated region), constant-temperature lines in the superheated region (which bend towards the horizontal as the pressure falls and the steam approaches ideal-gas behaviour, where h depends on T alone), constant-dryness-fraction lines in the wet region, constant-specific-volume lines, and the **saturation line $x = 1$** separating the two regions. The critical point lies far down on the left, the saturated liquid line is not usually shown, and only the region of engineering interest – x above about 0.6 and pressures from about 0.01 to 100 MPa – is drawn. A vertical line on the chart represents a reversible adiabatic (isentropic) process; a horizontal line represents throttling.

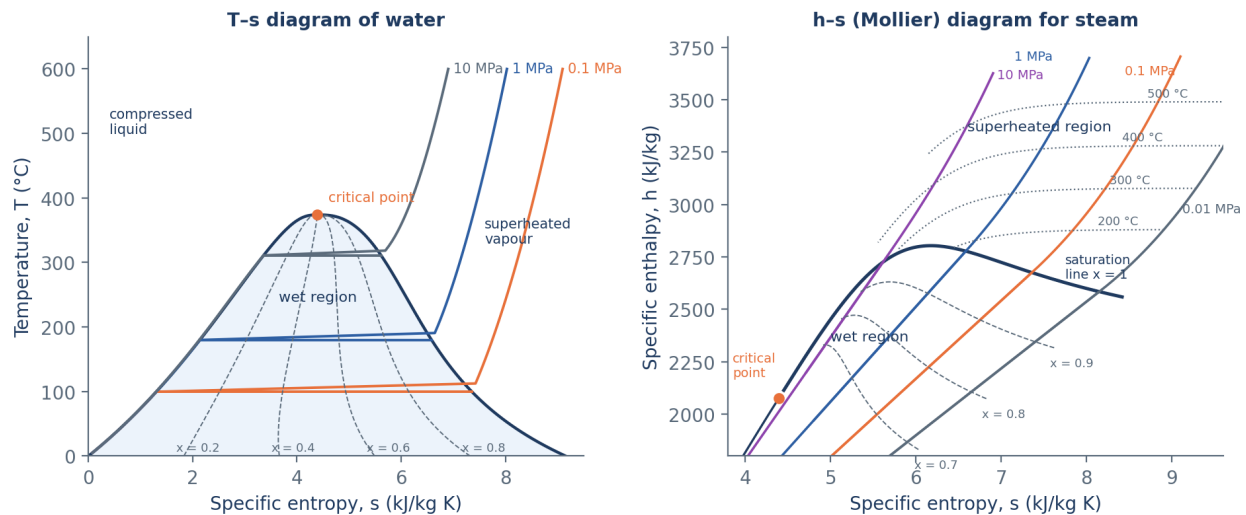


Figure 3.4 (Left) $T-s$ diagram of water with isobars and constant-dryness lines. (Right) The region of the $h-s$ (Mollier) diagram used in engineering calculations.

KEY POINT Reading the diagrams: inside the dome (wet region) p and T are *not* independent – an isobar and an isotherm are the same horizontal line – so a second property (v , h , s or x) is required to fix the state. Outside the dome (compressed liquid, superheated vapour) p and T are independent and fix the state on their own.

4. Steam tables and the properties of wet, dry saturated and superheated steam

4.1 The steam tables

The steam tables list the measured properties of water and steam at saturation and in the superheated region; the values in these notes are from the IAPWS-97 formulation, which the tables in the textbooks reproduce to within rounding.

- **Saturation table (temperature basis).** For each temperature: the saturation pressure and v_f , v_g , u_f , u_g , h_f , h_{fg} , h_g , s_f , s_{fg} , s_g . Used when the temperature of the wet steam or of the saturated state is known.
- **Saturation table (pressure basis).** The same properties listed against the pressure. Used when the pressure is known – the usual case in boilers, condensers and turbines.
- **Superheated steam table.** For each pressure, the properties v , u , h and s at temperatures above the saturation temperature, usually in steps of 50 K (the first row is the saturated vapour). Two independent properties, normally p and T , fix the state.
- **Compressed liquid table.** Properties of liquid water at pressures above the saturation pressure. Where it is not available the compressed liquid is treated as saturated liquid at the *same temperature*: $v \approx v_f(T)$, $u \approx u_f(T)$, $s \approx s_f(T)$ and $h \approx h_f(T) + v_f(p - p_{\text{sat}})$, the small correction term accounting for the flow work at the higher pressure.

Values between the tabulated entries are found by **linear interpolation**: if a property y is tabulated as y_1 at T_1 and y_2 at T_2 , then at T between them $y = y_1 + (y_2 - y_1)(T - T_1)/(T_2 - T_1)$. Interpolation is also used in reverse, to find the temperature at which a property has a given value (Worked Examples 3.2 and 3.14).

4.2 Properties of wet steam – the dryness fraction

Wet steam is a mixture of saturated liquid and saturated vapour at the same p and T_{sat} . Its composition is described by the **dryness fraction (quality)** x , the mass of dry vapour per unit mass of mixture:

$$x = m_g/(m_f + m_g) \quad 0 \leq x \leq 1 \quad \text{wetness fraction} = 1 - x = m_f/(m_f + m_g)$$

$x = 0$ is saturated liquid and $x = 1$ is dry saturated vapour; steam with $x = 0.9$ is "90 % dry" or "10 % wet". Consider 1 kg of wet steam containing x kg of vapour and $(1 - x)$ kg of liquid. The extensive properties – volume, internal energy, enthalpy and entropy – are additive, so the specific properties of the mixture are the mass-weighted averages of the saturated values:

$$v = (1 - x)v_f + x v_g = v_f + x v_{fg} \quad h = h_f + x h_{fg} \quad s = s_f + x s_{fg} \quad u = u_f + x u_{fg} = h - pv$$

where $v_{fg} = v_g - v_f$, $h_{fg} = h_g - h_f$ and so on. At low pressures v_f is negligible compared with v_g and $v \approx x v_g$; at high pressures the full expression must be used. **Conditions of validity:** the relations apply only in the two-phase region, $0 \leq x \leq 1$, where the mixture is at the saturation temperature for its pressure and the phases are in equilibrium; they cannot be used for compressed liquid (x has no meaning) or superheated vapour, for which the corresponding tables are used. Conversely, any one of the relations gives x when the mixture property is known – e.g. from a measured specific volume, $x = (v - v_f)/v_{fg}$ (Worked Examples 3.2 and 3.4).

KEY POINT To *identify* the state of steam at a given pressure, compare the given property with the saturated values: $v < v_f$ (or $T < T_{\text{sat}}$) → compressed liquid; $v_f \leq v \leq v_g$ → wet steam, find x ; $v > v_g$ (or $T > T_{\text{sat}}$) → superheated, use the superheat table. The same test works with h , s or u .

Two short examples. (a) *Enthalpy of wet steam at 0.6 MPa with $x = 0.85$:* from the tables $h_f = 670.5$ kJ/kg and $h_{fg} = 2085.6$ kJ/kg, so $h = 670.5 + 0.85 \times 2085.6 = 2443.3$ kJ/kg. (b) *Dryness fraction from a measured volume:* steam at 0.8 MPa has $v = 0.2$ m³/kg; with $v_f = 0.001115$ and $v_g = 0.24033$ m³/kg, $x = (0.2 - 0.001115)/(0.24033 - 0.001115) = 0.1989/0.2392 = 0.831$ (neglecting v_f , $x \approx 0.2/0.24033 = 0.832$).

4.3 Dry saturated and superheated steam

For dry saturated steam the properties are read directly: v_g , u_g , h_g , s_g at the given p or T . For superheated steam at (p, T) they are read from the superheat table; the internal energy, if not tabulated, is $u = h - pv$ (with p in kPa and v in m³/kg). When only the mean specific heat of superheated steam c_{ps} (about 2.0 to 2.1 kJ/kg K at low pressures) is available, the enthalpy may be approximated by $h_{\text{sup}} \approx h_g + c_{ps}(T_{\text{sup}} - T_{\text{sat}})$, but the table values should be used whenever they are available. The latent heat and the entropy of evaporation are linked by the fact that evaporation is a reversible constant-temperature process, $q = T ds$:

$$h_{fg} = T_{\text{sat}} s_{fg} \quad (T_{\text{sat}} \text{ in kelvin})$$

The **internal energy of steam** is its enthalpy less the flow (pressure–volume) energy, $u = h - pv$; for wet steam $u = (h_f + x h_{fg}) - p(v_f + x v_{fg}) = u_f + x u_{fg}$. It is the property needed for closed-system (non-flow) processes, whereas the enthalpy is the property needed for flow processes.

4.4 Steam as a real gas – comparison with the ideal-gas law

Superheated steam at low pressure and high superheat behaves approximately as an ideal gas with $R = 8.3145/18.015 = 0.4615$ kJ/kg K, but near the saturation line the intermolecular attractions reduce the volume below the ideal value, and the deviation grows with pressure; at the critical point the compressibility factor $Z = pv/RT$ is only 0.23. Worked Example 3.3 shows a 2 % deviation for steam at 1.2 MPa with 162 K of superheat; at 10 MPa and 350 °C the deviation is about 28 %. The steam tables must therefore be used for all but rough estimates at low pressure.

4.5 Worked examples on steam properties

WORKED EXAMPLE 3.1 — Saturation properties at three pressures and the latent heat–entropy relation

Problem. Using steam tables find, at 1.5 MPa, the saturation temperature, the change in specific volume and in entropy during evaporation, and the latent heat. Repeat at 0.15 MPa and at 8 MPa, verify in each case that $h_{fg} = T_{\text{sat}} s_{fg}$, and explain the trend of the quantities with pressure.

Given: Saturation table (pressure basis). At 1.5 MPa: $T_{\text{sat}} = 198.30^\circ\text{C}$, $v_f = 0.001154$, $v_g = 0.13170\text{ m}^3/\text{kg}$, $s_f = 2.3147$, $s_g = 6.4431\text{ kJ/kg K}$, $h_f = 844.7$, $h_g = 2791.0\text{ kJ/kg}$. At 0.15 MPa: $T_{\text{sat}} = 111.35^\circ\text{C}$, $v_f = 0.001053$, $v_g = 1.15936\text{ m}^3/\text{kg}$, $s_f = 1.4335$, $s_g = 7.2229\text{ kJ/kg K}$, $h_f = 467.1$, $h_g = 2693.1\text{ kJ/kg}$. At 8 MPa: $T_{\text{sat}} = 295.01^\circ\text{C}$, $v_f = 0.001385$, $v_g = 0.02353\text{ m}^3/\text{kg}$, $s_f = 3.2077$, $s_g = 5.7448\text{ kJ/kg K}$, $h_f = 1317.1$, $h_g = 2758.6\text{ kJ/kg}$.

Formula: $v_{fg} = v_g - v_f$; $s_{fg} = s_g - s_f$; $h_{fg} = h_g - h_f$; evaporation is reversible and isothermal, so $h_{fg} = q = T_{\text{sat}} s_{fg}$ with T_{sat} in kelvin.

Solution.

p (MPa)	$T_{\text{sat}} (^\circ\text{C})$	$T_{\text{sat}} (\text{K})$	$v_{fg} (\text{m}^3/\text{kg})$	$s_{fg} (\text{kJ/kg K})$	$h_{fg} (\text{kJ/kg})$	$T_{\text{sat}} s_{fg} (\text{kJ/kg})$
1.5	198.30	471.45	$0.13170 - 0.001154 = 0.13055$	$6.4431 - 2.3147 = 4.1284$	$2791.0 - 844.7 = 1946.3$	$471.45 \times 4.1284 = 1946.3$
0.15	111.35	384.50	$1.15936 - 0.001053 = 1.1583$	$7.2229 - 1.4335 = 5.7894$	$2693.1 - 467.1 = 2226.0$	$384.50 \times 5.7894 = 2226.0$
8	295.01	568.16	$0.02353 - 0.001385 = 0.02214$	$5.7448 - 3.2077 = 2.5372$	$2758.6 - 1317.1 = 1441.5$	$568.16 \times 2.5372 = 1441.5$

At 1.5 MPa: $T_{\text{sat}} = 198.3^\circ\text{C}$; $\Delta v = 0.13055\text{ m}^3/\text{kg}$; $\Delta s = 4.1284\text{ kJ/kg K}$; $h_{fg} = 1946.3\text{ kJ/kg}$. The last two columns agree at all three pressures, confirming $h_{fg} = T_{\text{sat}} s_{fg}$ ✓.

ANSWER With increasing pressure T_{sat} rises ($111.4 \rightarrow 198.3 \rightarrow 295.0^\circ\text{C}$) while v_{fg} ($1.158 \rightarrow 0.131 \rightarrow 0.022\text{ m}^3/\text{kg}$), s_{fg} ($5.79 \rightarrow 4.13 \rightarrow 2.54\text{ kJ/kg K}$) and h_{fg} ($2226 \rightarrow 1946 \rightarrow 1442\text{ kJ/kg}$) all fall, because the saturated liquid and saturated vapour states approach each other as the critical point (22.06 MPa) is approached, where $v_{fg} = s_{fg} = h_{fg} = 0$.

WORKED EXAMPLE 3.2 — Identifying the state of steam and constant-pressure heating

Problem. Steam at 1.5 MPa has a specific volume of $0.10\text{ m}^3/\text{kg}$. Identify its state and find its dryness fraction, enthalpy, entropy and internal energy. The steam is then heated at constant pressure until its specific volume is $0.14\text{ m}^3/\text{kg}$; find the final temperature and the heat supplied per kg.

Given: $p = 1.5\text{ MPa}$: $T_{\text{sat}} = 198.3^\circ\text{C}$, $v_f = 0.001154$, $v_g = 0.13170\text{ m}^3/\text{kg}$, $h_f = 844.7$, $h_g = 2791.0\text{ kJ/kg}$, $s_f = 2.3147$, $s_g = 6.4431\text{ kJ/kg K}$. Superheat table at 1.5 MPa: 200°C : $v = 0.13244$, $h = 2796.0$; 225°C : $v = 0.14258$, $h = 2862.9\text{ kJ/kg}$.

Formula: $x = (v - v_f)/(v_g - v_f)$; $h = h_f + x h_{fg}$; $s = s_f + x s_{fg}$; $u = h - pv$; constant-pressure heating (closed system): $q = h_2 - h_1$.

Solution.

Since $v_f < 0.10 < v_g$ the steam is **wet** at 198.3°C .

$x = (0.10 - 0.001154)/(0.13170 - 0.001154) = 0.098846/0.130546 = 0.757$.

$h_1 = 844.7 + 0.757 \times 1946.3 = 844.7 + 1473.5 = 2318.2 \text{ kJ/kg.}$
 $s_1 = 2.3147 + 0.757 \times 4.1284 = 2.3147 + 3.1252 = 5.440 \text{ kJ/kg K.}$
 $u_1 = h_1 - pv_1 = 2318.2 - 1500 \times 0.10 = 2318.2 - 150.0 = 2168.2 \text{ kJ/kg.}$
 Final state: $v_2 = 0.14 > v_g = 0.1317$, so the steam is superheated. Interpolating between 200°C and 225°C : $T_2 = 200 + 25 \times (0.14 - 0.13244)/(0.14258 - 0.13244) = 200 + 25 \times 0.7456 = 218.6^\circ\text{C}$; $h_2 = 2796.0 + 0.7456 \times (2862.9 - 2796.0) = 2796.0 + 49.9 = 2845.9 \text{ kJ/kg.}$
 $q = h_2 - h_1 = 2845.9 - 2318.2 = 527.7 \text{ kJ/kg.}$

ANSWER Wet steam, $x = 0.757$, $h = 2318 \text{ kJ/kg}$, $s = 5.440 \text{ kJ/kg K}$, $u = 2168 \text{ kJ/kg}$; after heating $T_2 = 218.6^\circ\text{C}$ (20.3 K of superheat) and $q = 527.7 \text{ kJ/kg}$.

WORKED EXAMPLE 3.3 — Superheated steam – table values and the ideal-gas comparison

Problem. Find the enthalpy, entropy, specific volume and internal energy of steam at 1.2 MPa and 350°C , and the degree of superheat. Compare the specific volume with that given by the ideal-gas equation with $R = 0.4615 \text{ kJ/kg K}$ and explain the difference.

Given: At 1.2 MPa: $T_{\text{sat}} = 188.0^\circ\text{C}$; superheat table at 1.2 MPa, 350°C : $v = 0.23454 \text{ m}^3/\text{kg}$, $h = 3154.1 \text{ kJ/kg}$, $s = 7.2138 \text{ kJ/kg K}$.

Formula: $u = h - pv$; degree of superheat $= T - T_{\text{sat}}$; ideal gas $v = RT/p$.

Solution.

Degree of superheat $= 350 - 188.0 = 162.0 \text{ K}$.

$u = 3154.1 - 1200 \times 0.23454 = 3154.1 - 281.4 = 2872.7 \text{ kJ/kg}$.

Ideal gas: $v = 0.4615 \times 623.15/1200 = 0.23965 \text{ m}^3/\text{kg}$.

Deviation $= (0.23965 - 0.23454)/0.23454 = +2.2\%$.

ANSWER $h = 3154.1 \text{ kJ/kg}$, $s = 7.2138 \text{ kJ/kg K}$, $v = 0.23454 \text{ m}^3/\text{kg}$, $u = 2872.7 \text{ kJ/kg}$, superheat 162 K. The ideal-gas volume is 2.2 % too large because the attractive forces between the molecules of the real vapour pull them closer together (compressibility factor $Z = pv/RT = 0.979$); the deviation would vanish at very low pressure and grow rapidly nearer the saturation line or at higher pressure.

WORKED EXAMPLE 3.4 — Liquid–vapour mixture in a vessel

Problem. A vessel of volume 0.05 m^3 contains a mixture of saturated water and saturated steam at 240°C ; the mass of liquid is 8 kg. Find (i) the pressure, (ii) the total mass and the dryness fraction, (iii) the specific volume, (iv) the enthalpy, (v) the entropy and (vi) the internal energy of the mixture.

Given: $V = 0.05 \text{ m}^3$; $m_f = 8 \text{ kg}$; at 240°C : $p_{\text{sat}} = 3.3467 \text{ MPa}$, $v_f = 0.001229$, $v_g = 0.05971 \text{ m}^3/\text{kg}$, $h_f = 1037.5$, $h_{fg} = 1765.5 \text{ kJ/kg}$, $s_f = 2.7019$, $s_{fg} = 3.4406 \text{ kJ/kg K}$.

Formula: $V_f = m_f v_f$; $V_g = V - V_f$; $m_g = V_g/v_g$; $x = m_g/(m_f + m_g)$; $v = V/m$; $h = h_f + x h_{fg}$; $s = s_f + x s_{fg}$; $u = h - pv$.

Solution.

(i) The mixture is saturated at 240°C , so $p = p_{\text{sat}} = 3.3467 \text{ MPa}$.

(ii) $V_f = 8 \times 0.001229 = 0.009832 \text{ m}^3$; $V_g = 0.05 - 0.009832 = 0.040168 \text{ m}^3$; $m_g = 0.040168/0.05971 = 0.6727 \text{ kg}$.

$m = 8 + 0.6727 = 8.6727 \text{ kg}$; $x = 0.6727/8.6727 = 0.0776$.

(iii) $v = 0.05/8.6727 = 0.005765 \text{ m}^3/\text{kg}$ (check: $v_f + x v_{fg} = 0.001229 + 0.0776 \times 0.05848 = 0.005766$ ✓).

(iv) $h = 1037.5 + 0.0776 \times 1765.5 = 1037.5 + 137.0 = 1174.5 \text{ kJ/kg}$.

(v) $s = 2.7019 + 0.0776 \times 3.4406 = 2.7019 + 0.2670 = 2.9689 \text{ kJ/kg K}$.
 (vi) $u = h - pv = 1174.5 - 3346.7 \times 0.005765 = 1174.5 - 19.3 = 1155.2 \text{ kJ/kg}$.

ANSWER $p = 3.347 \text{ MPa}$; $m = 8.673 \text{ kg}$, $x = 0.0776$; $v = 0.005765 \text{ m}^3/\text{kg}$; $h = 1174.5 \text{ kJ/kg}$; $s = 2.969 \text{ kJ/kg K}$; $u = 1155.2 \text{ kJ/kg}$. Although the vapour occupies 80 % of the volume it is less than 8 % of the mass – the quality of a mixture is a mass fraction, not a volume fraction.

5. Determination of the dryness fraction – calorimeters

The dryness fraction of steam leaving a boiler or flowing in a pipe cannot be measured directly; it is found from a **steam calorimeter** that brings a sample of the steam to a state whose properties can be measured and applies the energy balance. The instruments in use are the barrel (bucket) calorimeter, the separating calorimeter, the throttling calorimeter, the combined separating and throttling calorimeter and the electrical calorimeter (Figure 3.5). The sample is taken through a perforated tube inserted across the pipe so that it is representative of the flow.

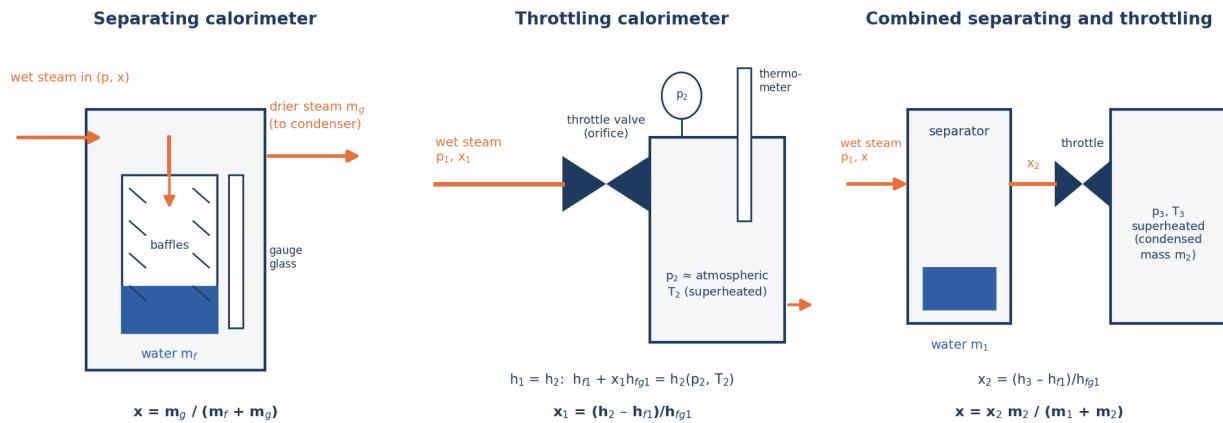


Figure 3.5 The separating calorimeter, the throttling calorimeter and the combined separating and throttling calorimeter.

5.1 Barrel (bucket) calorimeter

A known mass of wet steam m_s is condensed in a known mass of cold water m_w (initial temperature t_1) in an insulated barrel, and the final temperature t_2 is measured. The heat lost by the steam in condensing and cooling equals the heat gained by the water and the barrel (water equivalent m_e): $m_s[h_f + x h_{fg} - c_w t_2] = (m_w + m_e)c_w(t_2 - t_1)$, from which x is found. The method is simple but inaccurate, because the heat losses and the uncertainty in the small temperature rise are large compared with the quantity being measured.

5.2 Separating calorimeter

The wet steam is passed through a chamber in which sudden changes of direction (baffles) throw the heavier water droplets out of the stream by their inertia; the water collects at the bottom, where its mass m_f is read from a gauge glass, while the drier steam passes on to be condensed and weighed (m_g). Then

$$x = m_g / (m_f + m_g) \quad (\text{separating calorimeter})$$

The separation is never complete – some fine moisture is carried over with the steam – so the calorimeter over-estimates the dryness fraction. It is, however, the only instrument suitable for very wet steam, and it is used as the first stage of the combined calorimeter.

5.3 Throttling calorimeter

The sample is throttled through a partly open valve or orifice from the pipe pressure p_1 to a low pressure p_2 (a little above atmospheric) in an insulated chamber, where the pressure and temperature T_2 are measured. Throttling is a constant-enthalpy process (Unit I), and if the steam after throttling is *superheated* its state is fixed by p_2 and T_2 and its enthalpy h_2 is read from the superheat table. Equating the enthalpies before and after throttling,

$$h_{f1} + x_1 h_{fg1} = h_2(p_2, T_2) \quad \Rightarrow \quad x_1 = (h_2 - h_{f1})/h_{fg1}$$

On the Mollier chart the process is a horizontal line from the wet region at p_1 to the superheated region at p_2 . The calorimeter is accurate, but it works only if the throttled steam is superheated by at least about 5 K, which limits it to fairly dry steam (x above roughly 0.95 for the usual pressure ratios). For wetter steam the two instruments are combined.

5.4 Combined separating and throttling calorimeter

The steam first passes through the separating calorimeter, which removes most of the moisture (mass m_1 collected), and the partially dried steam (dryness x_2 , mass m_2 measured after condensation) is then throttled to p_3 , T_3 in the superheated region. The throttling calorimeter gives $x_2 = (h_3 - h_{f1})/h_{fg1}$, where h_{f1} and h_{fg1} are at the pipe pressure p_1 (the pressure is essentially unchanged in the separator). The dry vapour in the original sample is $x_2 m_2$ and the total mass of the sample is $m_1 + m_2$, so

$$x = x_2 m_2 / (m_1 + m_2) \quad (\text{combined calorimeter; } x = x_1 x_2 \text{ where } x_1 = m_2 / (m_1 + m_2) \text{ is the separator reading})$$

5.5 Worked examples on calorimeters

WORKED EXAMPLE 3.5 — Throttling calorimeter

Problem. Steam at 1.2 MPa is throttled in a throttling calorimeter to 0.1 MPa, where its temperature is 120 °C. Find the dryness fraction of the steam in the main. State the condition for the calorimeter to give a valid result.

Given: $p_1 = 1.2$ MPa: $h_{f1} = 798.5$ kJ/kg, $h_{fg1} = 1985.3$ kJ/kg; after throttling $p_2 = 0.1$ MPa, $T_2 = 120$ °C ($T_{\text{sat}} = 99.6$ °C, so the steam is superheated by 20.4 K): $h_2 = 2716.6$ kJ/kg.

Formula: Throttling: $h_1 = h_2$; $x_1 = (h_2 - h_{f1})/h_{fg1}$.

Solution.

$$x_1 = (2716.6 - 798.5)/1985.3 = 1918.1/1985.3 = 0.966.$$

ANSWER The steam in the main is 96.6 % dry. The result is valid because the throttled steam is superheated (120 °C > 99.6 °C), so that p_2 and T_2 fix h_2 ; had the steam been wetter, the throttled state would have remained wet and its enthalpy could not have been found from p_2 and T_2 alone.

WORKED EXAMPLE 3.6 — Combined separating and throttling calorimeter

Problem. In a test on a steam main at 1.0 MPa with a combined separating and throttling calorimeter, the separator collected 0.6 kg of water while 8 kg of steam passed through the throttle valve, leaving it at 0.11 MPa and 115 °C. Find the dryness fraction of the steam in the main.

Given: $p_1 = 1.0$ MPa: $h_{f1} = 762.7$ kJ/kg, $h_{fg1} = 2014.4$ kJ/kg; $m_1 = 0.6$ kg; $m_2 = 8$ kg; after throttling 0.11 MPa, 115 °C (superheated; $T_{\text{sat}} = 102.3$ °C): $h_3 = 2705.4$ kJ/kg.

Formula: $x_2 = (h_3 - h_{f1})/h_{fg1}$; $x = x_2 m_2 / (m_1 + m_2)$.

Solution.

Throttling stage: $x_2 = (2705.4 - 762.7)/2014.4 = 1942.7/2014.4 = 0.9644$.

Separating stage: the mass of dry vapour in the sample = $x_2 m_2 = 0.9644 \times 8 = 7.715$ kg; total sample mass = $0.6 + 8 = 8.6$ kg.

$x = 7.715/8.6 = 0.897$.

ANSWER The steam in the main is 89.7 % dry ($x_1 = 8/8.6 = 0.930$ from the separator alone, $x = 0.930 \times 0.9644 = 0.897$). The separator alone would have over-estimated the dryness as 0.930 because it does not remove the finest moisture, and the throttling calorimeter alone could not have been used on steam this wet.

6. Non-flow processes of steam – work and heat transfer

For a closed system (steam in a cylinder or a rigid vessel) the first law is $Q = \Delta U + W$, with $W = \int p \, dV$ for a quasi-static process. Unlike an ideal gas, steam has no simple equation of state, so the properties at the end states are taken from the tables and the process relations are used only to fix the end state and to evaluate the work. The standard processes are summarised below and illustrated in Figure 3.6.

Process	Condition	Work W_{1-2} (per kg)	Heat Q_{1-2} (per kg)	Remarks
Constant volume	$v_2 = v_1$	0	$u_2 - u_1$	Rigid vessel; on cooling, superheated steam becomes saturated when $v_g(T) = v$, then wet with $x = (v - v_f)/v_{fg}$
Constant pressure	$p_2 = p_1$	$p(v_2 - v_1)$	$h_2 - h_1$	Boilers, condensers; in the wet region also isothermal
Isothermal	$T_2 = T_1$	$q - (u_2 - u_1)$	$T(s_2 - s_1)$ (reversible)	In the wet region identical with constant pressure; in the superheated region p falls as v rises
Hyperbolic	$pv = C$	$p_1 v_1 \ln(v_2/v_1)$	$(u_2 - u_1) + W$	Not isothermal for steam (only for an ideal gas)
Isentropic (reversible adiabatic)	$s_2 = s_1$	$u_1 - u_2$	0	Ideal expansion in engines; x_2 from $s_2 = s_f + x_2 s_{fg}$
Polytropic	$pv^n = C$	$(p_1 v_1 - p_2 v_2)/(n - 1)$	$(u_2 - u_1) + W$	$n \approx 1.13$ for wet steam, 1.3 for superheated steam (Zeuner); v_2 from the process law, then x_2 or T_2 from the tables

For each process the procedure is: (1) fix state 1 from the tables; (2) use the process condition (v , p , T , s constant or $pv^n = C$) together with one given quantity to fix state 2 – deciding first whether state 2 is wet or superheated; (3) read u_2 , h_2 , s_2 ; (4) compute W from the table above and Q from the first law. For a mass m multiply by m .

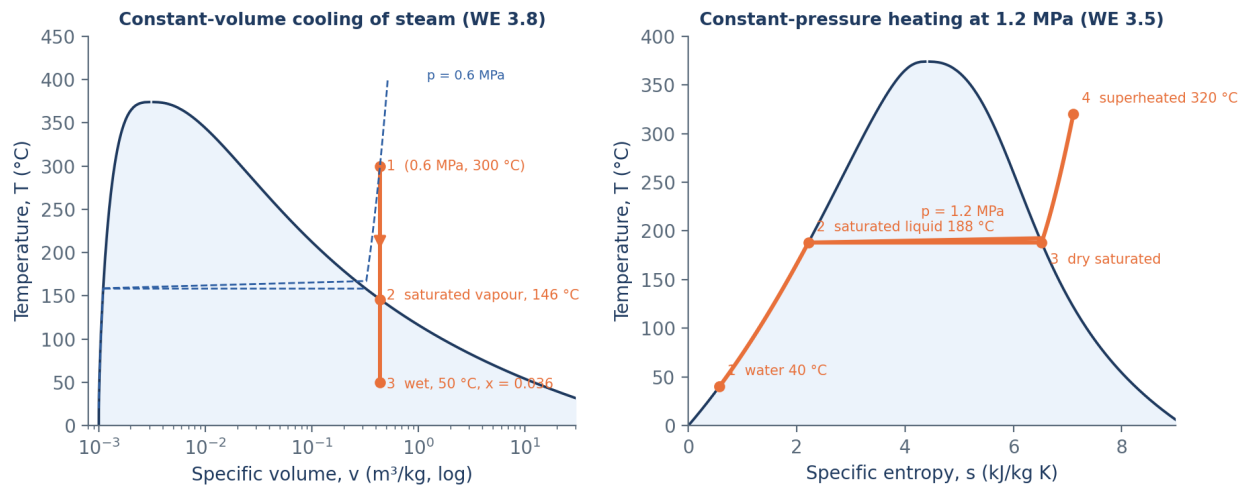


Figure 3.6 (Left) Constant-volume cooling of superheated steam on the T - v diagram (Worked Example 3.10): the steam becomes saturated where the vertical line meets the saturated vapour line and is wet below it. (Right) Constant-pressure heating of water from 40 °C to superheated steam at 320 °C at 1.2 MPa on the T - s diagram (Worked Example 3.7).

KEY POINT In a closed system the heat transfer is $Q = \Delta U + W$, and only for a *constant-pressure* process does it equal the enthalpy change ($Q = \Delta H$, because $W = p\Delta V$). For a rigid vessel $W = 0$ and $Q = \Delta U$, so the internal energy $u = h - pv$ must be used – a frequent source of error is to use $h_2 - h_1$ for a constant-volume process.

6.1 Worked examples on non-flow processes

WORKED EXAMPLE 3.7 — Constant-pressure heating of water to superheated steam – first-law check

Problem. A mass of 8 kg of water at 40 °C is heated at a constant pressure of 1.2 MPa until it becomes superheated steam at 320 °C. Show the process on the T - s and p - v diagrams, marking the saturated liquid and saturated vapour states crossed, and calculate the changes in volume, enthalpy, internal energy and entropy. Hence find the heat supplied and the work done, and verify that the first law is satisfied.

Given: $m = 8$ kg; $p = 1200$ kPa. State 1, compressed liquid at 40 °C (treated as saturated liquid at 40 °C with the enthalpy correction): $v_f = 0.001008$ m³/kg, $u_f = 167.5$ kJ/kg, $h_f = 167.5$ kJ/kg, $s_f = 0.5724$ kJ/kg K, $p_{\text{sat}}(40 \text{ °C}) = 7.38$ kPa. Saturated states at 1.2 MPa: $T_{\text{sat}} = 188.0$ °C, $v_f = 0.001139$, $v_g = 0.16325$ m³/kg, $h_f = 798.5$, $h_g = 2783.8$ kJ/kg, $s_f = 2.2163$, $s_g = 6.5217$ kJ/kg K. State 2, 1.2 MPa, 320 °C: $v = 0.22219$ m³/kg, $h = 3089.7$ kJ/kg, $s = 7.1078$ kJ/kg K.

Formula: $h_1 = h_f + v_f(p - p_{\text{sat}})$; $u_2 = h_2 - pv_2$; $\Delta V = m(v_2 - v_1)$; $W = p\Delta V$; $Q = \Delta H = m(h_2 - h_1)$; check $Q = \Delta U + W$.

Solution.

State 1: $h_1 = 167.5 + 0.001008 \times (1200 - 7.4) = 167.5 + 1.2 = 168.7$ kJ/kg; $u_1 = 167.5$ kJ/kg; $v_1 = 0.001008$ m³/kg; $s_1 = 0.5724$ kJ/kg K.

State 2: $u_2 = 3089.7 - 1200 \times 0.22219 = 3089.7 - 266.6 = 2823.1$ kJ/kg.

The process (Figure 3.6, right) passes through the saturated liquid state (188.0 °C, $h = 798.5$ kJ/kg) and the dry saturated state (188.0 °C, $h = 2783.8$ kJ/kg) before superheating; on the p - v diagram it is a horizontal line at 1.2 MPa from $v = 0.001008$ to 0.22219 m³/kg.

Change in volume: $\Delta V = 8 \times (0.22219 - 0.001008) = 8 \times 0.22118 = 1.769$ m³.

Change in enthalpy: $\Delta H = 8 \times (3089.7 - 168.7) = 8 \times 2921.0 = 23368$ kJ.

Change in internal energy: $\Delta U = 8 \times (2823.1 - 167.5) = 8 \times 2655.6 = 21245$ kJ.

Change in entropy: $\Delta S = 8 \times (7.1078 - 0.5724) = 8 \times 6.5354 = 52.28$ kJ/K.

Work done: $W = p\Delta V = 1200 \times 1.769 = 2123$ kJ.

Heat supplied (constant pressure): $Q = \Delta H = 23368 \text{ kJ}$.

First-law check: $\Delta U + W = 21245 + 2123 = 23368 \text{ kJ} = Q \checkmark$.

ANSWER $\Delta V = 1.769 \text{ m}^3$; $\Delta H = 23368 \text{ kJ}$; $\Delta U = 21245 \text{ kJ}$; $\Delta S = 52.28 \text{ kJ/K}$; $Q = 23368 \text{ kJ}$; $W = 2123 \text{ kJ}$. About 9 % of the heat supplied leaves again as work done in pushing the piston back; the remaining 91 % is stored as internal energy.

WORKED EXAMPLE 3.8 — Steam in a rigid vessel cooled to a lower pressure

Problem. A rigid vessel of 0.7 m^3 contains 1 kg of steam at 2.5 bar . Find the specific volume, temperature, dryness fraction, internal energy, enthalpy and entropy of the steam. The vessel is then cooled until the pressure is 1 bar ; show the process on the T - s diagram and find the final dryness fraction and the heat rejected.

Given: $V = 0.7 \text{ m}^3$, $m = 1 \text{ kg}$, so $v = 0.7 \text{ m}^3/\text{kg}$ (constant). At 2.5 bar (0.25 MPa): $T_{\text{sat}} = 127.4 \text{ }^\circ\text{C}$, $v_f = 0.001067$, $v_g = 0.7187 \text{ m}^3/\text{kg}$, $h_f = 535.4$, $h_{fg} = 2181.2 \text{ kJ/kg}$, $s_f = 1.6072$, $s_{fg} = 5.4452 \text{ kJ/kg K}$. At 1 bar (0.1 MPa): $T_{\text{sat}} = 99.6 \text{ }^\circ\text{C}$, $v_f = 0.001043$, $v_g = 1.6940 \text{ m}^3/\text{kg}$, $u_f = 417.3$, $u_g = 2505.5 \text{ kJ/kg}$, $s_f = 1.3026$, $s_{fg} = 6.0562 \text{ kJ/kg K}$.

Formula: $x = (v - v_f)/(v_g - v_f)$; $h = h_f + x h_{fg}$; $u = h - pv$; $s = s_f + x s_{fg}$; constant volume: $W = 0$, $Q = m(u_2 - u_1)$.

Solution.

State 1: $v_1 = 0.7 \text{ m}^3/\text{kg}$ lies between v_f and v_g , so the steam is wet at $T_1 = 127.4 \text{ }^\circ\text{C}$.

$x_1 = (0.7 - 0.001067)/(0.7187 - 0.001067) = 0.69893/0.71763 = 0.974$.

$h_1 = 535.4 + 0.974 \times 2181.2 = 535.4 + 2124.5 = 2659.9 \text{ kJ/kg}$.

$u_1 = h_1 - p_1 v_1 = 2659.9 - 250 \times 0.7 = 2659.9 - 175.0 = 2484.9 \text{ kJ/kg}$.

$s_1 = 1.6072 + 0.974 \times 5.4452 = 1.6072 + 5.3036 = 6.911 \text{ kJ/kg K}$.

State 2 ($v_2 = 0.7 \text{ m}^3/\text{kg}$, 1 bar): $x_2 = (0.7 - 0.001043)/(1.6940 - 0.001043) = 0.69896/1.69296 = 0.413$.

$u_2 = 417.3 + 0.413 \times (2505.5 - 417.3) = 417.3 + 862.4 = 1279.7 \text{ kJ/kg}$; $s_2 = 1.3026 + 0.413 \times 6.0562 = 3.804 \text{ kJ/kg K}$.

Heat rejected: $Q = m(u_2 - u_1) = 1 \times (1279.7 - 2484.9) = -1205.2 \text{ kJ}$.

On the T - s diagram the process is a curve of constant specific volume running down and to the left inside the dome, from $(127.4 \text{ }^\circ\text{C}, 6.911)$ to $(99.6 \text{ }^\circ\text{C}, 3.804 \text{ kJ/kg K})$; the area under it is the heat rejected.

ANSWER $v = 0.7 \text{ m}^3/\text{kg}$; $T = 127.4 \text{ }^\circ\text{C}$; $x = 0.974$; $u = 2484.9 \text{ kJ/kg}$; $h = 2659.9 \text{ kJ/kg}$; $s = 6.911 \text{ kJ/kg K}$. After cooling to 1 bar $x_2 = 0.413$ and 1205 kJ of heat is rejected. Because the volume is fixed, more than half of the steam condenses as the pressure falls to 1 bar .

WORKED EXAMPLE 3.9 — Constant-volume cooling of superheated steam

Problem. Steam at 0.35 MPa and $250 \text{ }^\circ\text{C}$ is cooled at constant volume. Find (i) the temperature at which the steam becomes saturated vapour, (ii) the dryness fraction at $75 \text{ }^\circ\text{C}$ and (iii) the heat transferred per kg in cooling from $250 \text{ }^\circ\text{C}$ to $75 \text{ }^\circ\text{C}$.

Given: State 1, 0.35 MPa , $250 \text{ }^\circ\text{C}$ (superheated): $v_1 = 0.68145 \text{ m}^3/\text{kg}$, $u_1 = 2727.7 \text{ kJ/kg}$. Saturation table: at $125 \text{ }^\circ\text{C}$ $v_g = 0.77011$, at $130 \text{ }^\circ\text{C}$ $v_g = 0.66808 \text{ m}^3/\text{kg}$. At $75 \text{ }^\circ\text{C}$: $v_f = 0.001026$, $v_g = 4.1291 \text{ m}^3/\text{kg}$, $u_f = 313.9$, $u_g = 2475.2 \text{ kJ/kg}$.

Formula: Saturation is reached when $v_g(T) = v_1$ (interpolate); $x = (v_1 - v_f)/(v_g - v_f)$; $u_2 = u_f + x(u_g - u_f)$; constant volume: $q = u_2 - u_1$.

Solution.

(i) $v_1 = 0.68145 \text{ m}^3/\text{kg}$ lies between $v_g(125^\circ\text{C}) = 0.77011$ and $v_g(130^\circ\text{C}) = 0.66808$: $T_{\text{sat}} = 125 + 5 \times (0.77011 - 0.68145)/(0.77011 - 0.66808) = 125 + 5 \times 0.869 = 129.3^\circ\text{C}$ ($p \approx 0.265 \text{ MPa}$). Above this temperature the steam is superheated; below it, wet.

(ii) At 75°C : $x = (0.68145 - 0.001026)/(4.1291 - 0.001026) = 0.68042/4.1281 = 0.165$.

(iii) $u_2 = 313.9 + 0.165 \times (2475.2 - 313.9) = 313.9 + 356.6 = 670.5 \text{ kJ/kg}$.

$q = u_2 - u_1 = 670.5 - 2727.7 = -2057.2 \text{ kJ/kg}$.

ANSWER (i) The steam becomes dry saturated at 129.3°C ; (ii) $x = 0.165$ at 75°C ; (iii) 2057 kJ/kg of heat is rejected. The pressure at 75°C is only 38.6 kPa – cooling a closed vessel of steam produces a partial vacuum, which is how a steam condenser works.

WORKED EXAMPLE 3.10 — Rigid vessel cooled from superheated to wet – where does the heat go?

Problem. Steam at 0.6 MPa and 300°C in a rigid vessel is cooled to 50°C . Show the process on the T - v and p - v diagrams and find (i) the temperature and pressure at which condensation begins, (ii) the dryness fraction at 50°C and (iii) the heat rejected per kg. Compare the heat rejected while the steam is superheated with that rejected after condensation begins.

Given: State 1, 0.6 MPa , 300°C : $v_1 = 0.43441 \text{ m}^3/\text{kg}$, $u_1 = 2801.4 \text{ kJ/kg}$. Saturation table: at 145°C $v_g = 0.44602 \text{ m}^3/\text{kg}$, $p_{\text{sat}} = 415.6 \text{ kPa}$, $u_g = 2554.4 \text{ kJ/kg}$; at 150°C $v_g = 0.39250 \text{ m}^3/\text{kg}$, $p_{\text{sat}} = 476.1 \text{ kPa}$, $u_g = 2559.0 \text{ kJ/kg}$. At 50°C : $p_{\text{sat}} = 12.35 \text{ kPa}$, $v_f = 0.001012$, $v_g = 12.028 \text{ m}^3/\text{kg}$, $u_f = 209.3$, $u_g = 2442.8 \text{ kJ/kg}$.

Formula: Condensation begins where $v_g(T) = v_1$; $x_3 = (v_1 - v_f)/(v_g - v_f)$; $u = u_f + x u_{fg}$; $q = \Delta u$ for each stage.

Solution.

(i) Interpolating for $v_g = 0.43441$: $T_2 = 145 + 5 \times (0.44602 - 0.43441)/(0.44602 - 0.39250) = 145 + 5 \times 0.217 = 146.1^\circ\text{C}$; $p_2 = 415.6 + 0.217 \times (476.1 - 415.6) = 428.7 \text{ kPa}$; $u_2 = u_g = 2554.4 + 0.217 \times 4.6 = 2555.4 \text{ kJ/kg}$.

(ii) At 50°C : $x_3 = (0.43441 - 0.001012)/(12.028 - 0.001012) = 0.43340/12.027 = 0.0360$.

$u_3 = 209.3 + 0.0360 \times (2442.8 - 209.3) = 209.3 + 80.4 = 289.7 \text{ kJ/kg}$.

(iii) Superheated stage 1–2: $q_{1-2} = u_2 - u_1 = 2555.4 - 2801.4 = -246.0 \text{ kJ/kg}$.

Condensing stage 2–3: $q_{2-3} = u_3 - u_2 = 289.7 - 2555.4 = -2265.7 \text{ kJ/kg}$.

Total: $q = -246.0 - 2265.7 = -2511.7 \text{ kJ/kg}$ ($= u_3 - u_1 = 289.7 - 2801.4 \checkmark$).

ANSWER (i) Condensation begins at 146.1°C and 429 kPa ; (ii) $x = 0.036$ at 50°C ; (iii) 2512 kJ/kg is rejected in all. Only about 10 % of the heat (246 kJ/kg) is removed in cooling the superheated vapour through 154 K , while 90 % (2266 kJ/kg) is removed in cooling through the next 96 K , because in the wet region the heat removed is mostly the latent heat of the 96.4 % of the steam that condenses, not sensible heat. On the T - v diagram (Figure 3.6, left) the process is a vertical line; on the p - v diagram it is also vertical, the pressure falling from 600 kPa to 12.35 kPa .

7. Flow processes of steam – turbine, nozzle, throttling and mixing

Steam in power plants flows steadily through boilers, turbines, nozzles, condensers, valves and mixing chambers. Each device is analysed with the steady flow energy equation of Unit I, $h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$, with the enthalpies taken from the steam tables or the Mollier chart; the potential-energy term is always negligible. The property that governs flow processes is the **enthalpy** (not the internal energy), and the ideal (reversible adiabatic) process is **isentropic** – a vertical line on the h - s chart.

7.1 Steam turbine

The steam expands from the inlet state (p_1, T_1) to the exhaust pressure p_2 ; the casing is insulated ($q \approx 0$) and the velocities are small, so $w = h_1 - h_2$. For the ideal turbine the expansion is isentropic, $s_{2s} = s_1$; if s_1 lies between s_f and s_g at p_2 the exhaust is wet, with $x_{2s} = (s_1 - s_{f2})/s_{fg2}$ and $h_{2s} = h_{f2} + x_{2s}h_{fg2}$, and the ideal work is $w_s = h_1 - h_{2s}$ – the vertical drop on the Mollier chart. A real turbine has friction and turbulence, so the exhaust entropy is higher, the exhaust state lies to the right of $2s$ on the chart (Figure 3.7, left) and the work is smaller. The **isentropic (adiabatic) efficiency** compares the two:

$$\eta_T = (\text{actual work})/(\text{isentropic work}) = (h_1 - h_2)/(h_1 - h_{2s}) \quad h_2 = h_1 - \eta_T(h_1 - h_{2s})$$

From h_2 the actual exit dryness follows, $x_2 = (h_2 - h_{f2})/h_{fg2}$; it is *higher* than x_{2s} because the energy not converted into work remains in the steam as enthalpy and re-evaporates some of the moisture. The entropy generated per kg is $s_{\text{gen}} = s_2 - s_1$ (adiabatic), which the second law requires to be positive (Worked Example 3.12).

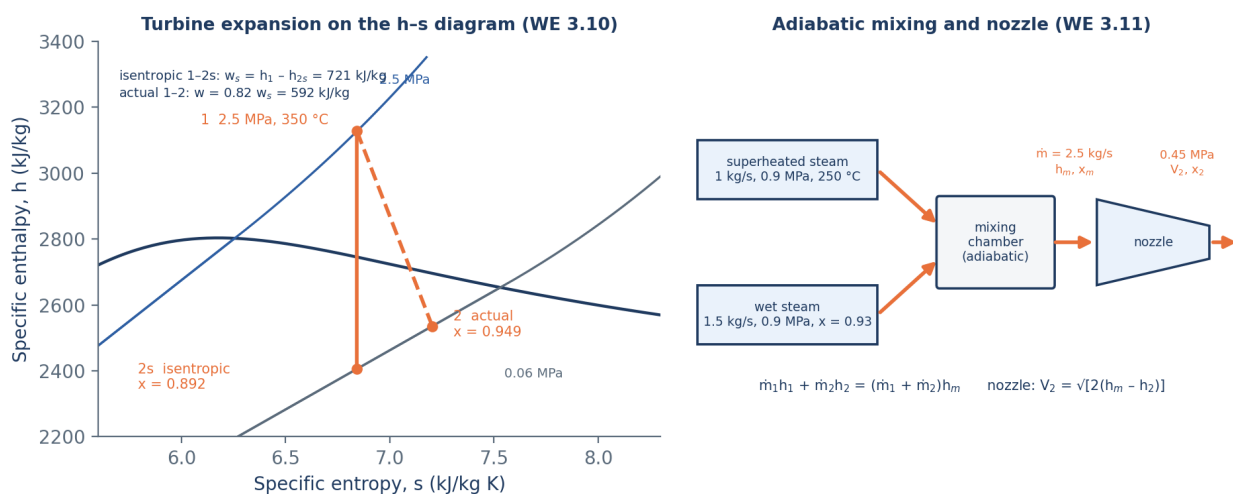


Figure 3.7 (Left) Isentropic and actual expansion of steam in a turbine on the h - s diagram (Worked Example 3.12). (Right) Adiabatic mixing of two steam streams followed by expansion in a nozzle (Worked Example 3.13).

7.2 Steam nozzle

In a nozzle the enthalpy drop is converted into kinetic energy; there is no work, no heat transfer and no change of elevation, so $V_2^2/2 - V_1^2/2 = h_1 - h_2$. For an ideal (frictionless) nozzle the expansion is isentropic and, with a negligible inlet velocity,

$$V_2 = \sqrt{2(h_1 - h_{2s})} \quad (h \text{ in J/kg}) = 44.72 \sqrt{(h_1 - h_{2s})} \quad (h \text{ in kJ/kg, } V \text{ in m/s}) \quad A_2 = \dot{m} v_2/V_2$$

The exit area follows from continuity with $v_2 = v_{f2} + x_2 v_{fg2}$. The velocity coefficient or nozzle efficiency, $\eta_N = (h_1 - h_2)/(h_1 - h_{2s})$, accounts for friction in real nozzles (Worked Examples 3.13 and 3.15).

7.3 Throttling

Steam flowing through a partly open valve, an orifice or a porous plug suffers a large pressure drop with no work, no heat transfer and negligible change of velocity, so $h_1 = h_2$ – a horizontal line on the Mollier chart, along which wet steam becomes drier and eventually superheated, and superheated steam becomes cooler (steam is not an ideal gas, so its temperature falls on throttling). Throttling is highly irreversible: the entropy increases ($s_2 > s_1$) and the pressure drop is lost. It is used in the throttling calorimeter (Section 5.3), in pressure-reducing valves, in the expansion valve of refrigerators (Unit V) and in throttle governing of turbines.

7.4 Adiabatic mixing of steam streams

When two streams at the same pressure mix in an insulated chamber with no work, the mass and energy balances give

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3 \quad \dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \quad \Rightarrow \quad h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2) / (\dot{m}_1 + \dot{m}_2)$$

The mixture state is fixed by p and h_3 : if $h_3 < h_g$ at that pressure it is wet, with $x_3 = (h_3 - h_f)/h_{fg}$; if $h_3 > h_g$ it is superheated and the temperature is found by interpolation in the superheat table. Mixing of streams in different states is irreversible; the rate of entropy generation $\dot{S}_{\text{gen}} = \dot{m}_3 s_3 - \dot{m}_1 s_1 - \dot{m}_2 s_2$ must be positive for the process to be possible (Worked Example 3.14). Desuperheaters, which spray water into superheated steam to bring it to saturation, and feed-water heaters work on this principle.

7.5 Worked examples on flow processes

WORKED EXAMPLE 3.11 — Isentropic expansion in a turbine to a given exhaust temperature

Problem. Steam at 1.4 MPa and 300 °C expands reversibly and adiabatically in a turbine to 45 °C. Sketch the process on the T – s and h – s diagrams and find (i) the exhaust pressure and dryness fraction and (ii) the ideal work per kg of steam.

Given: State 1, 1.4 MPa, 300 °C: $h_1 = 3041.0$ kJ/kg, $s_1 = 6.9553$ kJ/kg K. Saturation at 45 °C: $p_{\text{sat}} = 9.59$ kPa, $h_f = 188.4$, $h_{fg} = 2394.0$ kJ/kg, $s_f = 0.6386$, $s_{fg} = 7.5248$ kJ/kg K.

Formula: Isentropic: $s_2 = s_1$; $x_2 = (s_1 - s_f)/s_{fg}$; $h_2 = h_f + x_2 h_{fg}$; $w = h_1 - h_2$.

Solution.

(i) The exhaust is saturated at 45 °C, so $p_2 = p_{\text{sat}}(45 \text{ °C}) = 9.59$ kPa (a condenser vacuum). $s_1 = 6.9553$ lies between s_f and $s_g = 8.1634$ at 45 °C, so the exhaust is wet: $x_2 = (6.9553 - 0.6386)/7.5248 = 6.3167/7.5248 = 0.839$.

(ii) $h_2 = 188.4 + 0.839 \times 2394.0 = 188.4 + 2008.6 = 2197.0$ kJ/kg.

$w = h_1 - h_2 = 3041.0 - 2197.0 = 844.0$ kJ/kg.

On the T – s diagram the process is a vertical line from the superheated region at 1.4 MPa (300 °C) down into the wet region at 45 °C; on the h – s chart it is a vertical drop of 844 kJ/kg from the 1.4 MPa isobar to the 9.59 kPa isobar at $x = 0.839$.

ANSWER (i) $p_2 = 9.59$ kPa, $x_2 = 0.839$; (ii) ideal work = 844 kJ/kg. The exhaust is 16 % wet – more than the 10–12 % usually tolerated in turbine blading, which is why reheating is used in practice.

WORKED EXAMPLE 3.12 — Turbine with an isentropic efficiency – actual work and entropy generation

Problem. Steam at 2.5 MPa and 350 °C expands in a turbine to 0.06 MPa. (i) Find the exit dryness fraction and the work per kg for a reversible adiabatic expansion. (ii) If the isentropic efficiency is 82 %, find the actual work, the actual exit dryness fraction and the entropy generated per kg, and explain why the steam leaving the actual turbine is drier.

Given: State 1, 2.5 MPa, 350 °C: $h_1 = 3127.0$ kJ/kg, $s_1 = 6.8424$ kJ/kg K. At 0.06 MPa: $T_{\text{sat}} = 85.9$ °C, $h_f = 359.8$, $h_{fg} = 2293.0$ kJ/kg, $s_f = 1.1452$, $s_{fg} = 6.3859$ kJ/kg K; $\eta_T = 0.82$.

Formula: $x_{2s} = (s_1 - s_f)/s_{fg}$; $h_{2s} = h_f + x_{2s} h_{fg}$; $w_s = h_1 - h_{2s}$; $w = \eta_T w_s$; $h_2 = h_1 - w$; $x_2 = (h_2 - h_f)/h_{fg}$; $s_2 = s_f + x_2 s_{fg}$; $S_{\text{gen}} = S_2 - S_1$.

Solution.

(i) $x_{2s} = (6.8424 - 1.1452)/6.3859 = 5.6972/6.3859 = 0.892$.

$h_{2s} = 359.8 + 0.892 \times 2293.0 = 359.8 + 2045.4 = 2405.2$ kJ/kg.

$w_s = 3127.0 - 2405.2 = 721.8$ kJ/kg.

(ii) $w = 0.82 \times 721.8 = 591.9 \text{ kJ/kg}$; $h_2 = 3127.0 - 591.9 = 2535.1 \text{ kJ/kg}$.
 $x_2 = (2535.1 - 359.8)/2293.0 = 2175.3/2293.0 = 0.949$.
 $s_2 = 1.1452 + 0.949 \times 6.3859 = 1.1452 + 6.0602 = 7.2054 \text{ kJ/kg K}$.
 $s_{\text{gen}} = s_2 - s_1 = 7.2054 - 6.8424 = 0.363 \text{ kJ/kg K}$ (> 0 , so the process is possible).

ANSWER (i) $x_{2s} = 0.892$, $w_s = 722 \text{ kJ/kg}$; (ii) $w = 592 \text{ kJ/kg}$, $x_2 = 0.949$, $s_{\text{gen}} = 0.363 \text{ kJ/kg K}$. The actual exhaust is drier because the 130 kJ/kg of enthalpy drop not converted into work is dissipated by friction inside the steam, reheating it and evaporating part of the moisture; the price is the entropy generated – with surroundings at 25 °C the irreversibility is $T_0 s_{\text{gen}} = 298.15 \times 0.363 = 108 \text{ kJ/kg}$, which is the work lost ($721.8 - 591.9 = 130 \text{ kJ/kg}$ less the 22 kJ/kg by which the exergy of the drier exhaust exceeds that of the isentropic exhaust).

WORKED EXAMPLE 3.13 — Mixing of superheated and wet steam followed by a nozzle

Problem. Superheated steam at 0.9 MPa and 250 °C flowing at 1 kg/s enters a pipe carrying wet steam at 0.9 MPa and 0.93 dry; after adiabatic mixing the flow rate is 2.5 kg/s. Find the condition of the steam after mixing. The mixture then expands isentropically in a frictionless nozzle to 0.45 MPa; find the exit dryness fraction and velocity, neglecting the velocity in the pipe.

Given: $\dot{m}_1 = 1 \text{ kg/s}$ at 0.9 MPa, 250 °C: $h_1 = 2946.9 \text{ kJ/kg}$; $\dot{m}_2 = 1.5 \text{ kg/s}$ at 0.9 MPa, $x_2 = 0.93$; at 0.9 MPa: $h_f = 742.7$, $h_{fg} = 2030.3$, $h_g = 2773.0 \text{ kJ/kg}$, $s_f = 2.0944$, $s_{fg} = 4.5268 \text{ kJ/kg K}$. At 0.45 MPa: $h_f = 623.2$, $h_{fg} = 2120.2 \text{ kJ/kg}$, $s_f = 1.8206$, $s_{fg} = 5.0353 \text{ kJ/kg K}$.

Formula: $h_2 = h_f + x_2 h_{fg}$; $h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2)/(\dot{m}_1 + \dot{m}_2)$; $x_3 = (h_3 - h_f)/h_{fg}$; $s_3 = s_f + x_3 s_{fg}$; nozzle: $s_4 = s_3$, $x_4 = (s_3 - s_{f4})/s_{fg4}$, $h_4 = h_{f4} + x_4 h_{fg4}$, $V_4 = \sqrt{2(h_3 - h_4)}$.

Solution.

Wet stream: $h_2 = 742.7 + 0.93 \times 2030.3 = 742.7 + 1888.2 = 2630.9 \text{ kJ/kg}$.

Mixing: $h_3 = (1 \times 2946.9 + 1.5 \times 2630.9)/2.5 = (2946.9 + 3946.4)/2.5 = 2757.3 \text{ kJ/kg}$.

Since $h_3 < h_g = 2773.0 \text{ kJ/kg}$ the mixture is still wet: $x_3 = (2757.3 - 742.7)/2030.3 = 2014.6/2030.3 = 0.992$.

$s_3 = 2.0944 + 0.992 \times 4.5268 = 2.0944 + 4.4906 = 6.585 \text{ kJ/kg K}$.

Nozzle (isentropic to 0.45 MPa): $x_4 = (6.585 - 1.8206)/5.0353 = 4.7644/5.0353 = 0.946$.

$h_4 = 623.2 + 0.946 \times 2120.2 = 623.2 + 2005.7 = 2628.9 \text{ kJ/kg}$.

$V_4 = \sqrt{2 \times (2757.3 - 2628.9) \times 1000} = \sqrt{256800} = 507 \text{ m/s}$.

ANSWER After mixing the steam is wet, $x_3 = 0.992$ at 0.9 MPa (175.4 °C); at the nozzle exit $x_4 = 0.946$ and $V_4 = 507 \text{ m/s}$. The superheated steam has dried the wet stream almost completely – this is the principle of the desuperheater in reverse.

WORKED EXAMPLE 3.14 — Adiabatic mixing in a pipeline – possibility and entropy generation

Problem. Superheated steam at 1.2 MPa and 300 °C flowing at 2 kg/s is mixed adiabatically in a pipeline with wet steam at 1.2 MPa and 0.92 dry flowing at 1.2 kg/s. Find the enthalpy and temperature of the mixture and the rate of entropy generation, and state whether the process is possible.

Given: Stream 1: $\dot{m}_1 = 2 \text{ kg/s}$, $h_1 = 3046.4 \text{ kJ/kg}$, $s_1 = 7.0336 \text{ kJ/kg K}$. Stream 2: $\dot{m}_2 = 1.2 \text{ kg/s}$, $x_2 = 0.92$; at 1.2 MPa: $T_{\text{sat}} = 188.0 \text{ °C}$, $h_f = 798.5$, $h_{fg} = 1985.3$, $h_g = 2783.8 \text{ kJ/kg}$, $s_f = 2.2163$, $s_{fg} = 4.3054 \text{ kJ/kg K}$. Superheat table at 1.2 MPa: 200 °C: $h = 2816.1$, $s = 6.5908$; 250 °C: $h = 2935.7$, $s = 6.8314$.

Formula: $h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2)/(\dot{m}_1 + \dot{m}_2)$; T_3 and s_3 by interpolation; $\dot{S}_{\text{gen}} = \dot{m}_3 s_3 - \dot{m}_1 s_1 - \dot{m}_2 s_2 \geq 0$.

Solution.

Wet stream: $h_2 = 798.5 + 0.92 \times 1985.3 = 798.5 + 1826.5 = 2625.0 \text{ kJ/kg}$; $s_2 = 2.2163 + 0.92 \times$

$$4.3054 = 2.2163 + 3.9610 = 6.1773 \text{ kJ/kg K.}$$

$$h_3 = (2 \times 3046.4 + 1.2 \times 2625.0)/3.2 = (6092.8 + 3150.0)/3.2 = 2888.4 \text{ kJ/kg.}$$

$h_3 > h_g = 2783.8$, so the mixture is superheated. Interpolating between 200 °C and 250 °C: fraction = $(2888.4 - 2816.1)/(2935.7 - 2816.1) = 72.3/119.6 = 0.604$; $T_3 = 200 + 0.604 \times 50 = 230.2$ °C; $s_3 = 6.5908 + 0.604 \times (6.8314 - 6.5908) = 6.5908 + 0.1454 = 6.7362 \text{ kJ/kg K.}$

$$\dot{S}_{\text{gen}} = 3.2 \times 6.7362 - 2 \times 7.0336 - 1.2 \times 6.1773 = 21.556 - 14.067 - 7.413 = +0.076 \text{ kW/K.}$$

ANSWER $h_3 = 2888 \text{ kJ/kg}$; the mixture is superheated at 230 °C (42 K of superheat); $\dot{S}_{\text{gen}} = 0.076 \text{ kW/K}$. Since the entropy generation is positive the process is possible, and irreversible – the irreversibility with surroundings at 25 °C is $T_0 \dot{S}_{\text{gen}} = 298.15 \times 0.076 = 22.7 \text{ kW}$, the work potential destroyed by mixing streams in different states.

WORKED EXAMPLE 3.15 — Steam nozzle – exit dryness, velocity and area

Problem. Steam at 1.2 MPa and 250 °C expands isentropically through a frictionless nozzle to 0.35 MPa. Neglecting the inlet velocity, find (i) the dryness fraction at exit, (ii) the exit velocity and (iii) the exit area for a flow of 0.6 kg/s.

Given: State 1, 1.2 MPa, 250 °C: $h_1 = 2935.7 \text{ kJ/kg}$, $s_1 = 6.8314 \text{ kJ/kg K}$. At 0.35 MPa: $v_f = 0.001079$, $v_g = 0.5242 \text{ m}^3/\text{kg}$, $h_f = 584.3$, $h_{fg} = 2147.7 \text{ kJ/kg}$, $s_f = 1.7275$, $s_{fg} = 5.2126 \text{ kJ/kg K}$; $\dot{m} = 0.6 \text{ kg/s}$.

Formula: $x_2 = (s_1 - s_f)/s_{fg}$; $h_2 = h_f + x_2 h_{fg}$; $v_2 = v_f + x_2(v_g - v_f)$; $V_2 = \sqrt{2(h_1 - h_2)}$; $A_2 = \dot{m} v_2 / V_2$.

Solution.

$$(i) x_2 = (6.8314 - 1.7275)/5.2126 = 5.1039/5.2126 = 0.979.$$

$$(ii) h_2 = 584.3 + 0.979 \times 2147.7 = 584.3 + 2102.6 = 2686.9 \text{ kJ/kg.}$$

$$V_2 = \sqrt{2 \times (2935.7 - 2686.9) \times 1000} = \sqrt{497600} = 705 \text{ m/s.}$$

$$(iii) v_2 = 0.001079 + 0.979 \times (0.5242 - 0.001079) = 0.001079 + 0.5121 = 0.5132 \text{ m}^3/\text{kg.}$$

$$A_2 = 0.6 \times 0.5132/705 = 4.37 \times 10^{-4} \text{ m}^2 = 4.37 \text{ cm}^2.$$

ANSWER (i) $x_2 = 0.979$; (ii) $V_2 = 705 \text{ m/s}$; (iii) $A_2 = 4.37 \text{ cm}^2$. An enthalpy drop of only 249 kJ/kg produces a supersonic velocity of 705 m/s – which is why nozzles, not pistons, are used to convert the energy of steam into velocity in turbines.

8. Application of the first and second laws to pure substances

The two laws are applied to steam exactly as to any other substance; only the way the properties are obtained differs – from the tables and charts instead of from $p v = RT$.

First law – closed systems (non-flow). $Q = m(u_2 - u_1) + W$, with $W = \int p dV$ evaluated from the process law (Section 6). The internal energy is $u = h - p v$, or $u_f + x u_{fg}$ for wet steam. Constant volume: $Q = m \Delta u$; constant pressure: $Q = m \Delta h$; isentropic: $W = -m \Delta u$.

First law – open systems (steady flow). $q - w = \Delta h + \Delta V^2/2$ (Section 7): turbine $w = h_1 - h_2$, nozzle $V_2 = \sqrt{2(h_1 - h_2)}$, throttling $h_1 = h_2$, boiler and condenser $q = h_2 - h_1$, mixing $\sum \dot{m}_{\text{in}} h_{\text{in}} = \sum \dot{m}_{\text{out}} h_{\text{out}}$.

Second law – entropy of steam. The entropy of steam is read from the tables ($s = s_f + x s_{fg}$ in the wet region) so that the entropy change of any process is $s_2 - s_1$ regardless of the path. For a reversible process the heat transfer is the area under the T – s curve, $q = \int T ds$ – for the isothermal evaporation $q = T_{\text{sat}} s_{fg} = h_{fg}$. A reversible adiabatic process is isentropic, and the isentropic end state is the reference for the actual one through the isentropic efficiency of turbines, compressors and nozzles.

Second law – entropy generation and feasibility. For an adiabatic device $s_{\text{gen}} = s_{\text{out}} - s_{\text{in}}$ per kg (turbine, nozzle, throttle) or $\dot{S}_{\text{gen}} = \sum \dot{m}_{\text{out}} s_{\text{out}} - \sum \dot{m}_{\text{in}} s_{\text{in}}$ (mixing); for a device exchanging heat q with surroundings at

T_0 , $s_{\text{gen}} = (s_2 - s_1) - q/T_0$. The process is possible only if $s_{\text{gen}} \geq 0$, and the irreversibility (work lost) is $I = T_0 s_{\text{gen}}$ (Unit II). The examples of this unit illustrate each case: the actual turbine of Worked Example 3.12 generates 0.363 kJ/kg K; the mixing of Worked Example 3.14 generates 0.076 kW/K; throttling in the calorimeter of Worked Example 3.5 raises the entropy from $s_1 = 2.2163 + 0.966 \times 4.3054 = 6.375$ to $s_2 = 7.468$ kJ/kg K, an increase of 1.09 kJ/kg K – the largest of the three, showing how wasteful throttling is as a means of reducing pressure.

Availability of steam. The availability of a stream of steam, $\psi = (h - h_0) - T_0(s - s_0)$ with the dead state usually taken as saturated liquid at the atmospheric temperature, gives the maximum work obtainable from it; the second-law efficiency of a steam turbine is $\eta_{\text{II}} = w/(\psi_1 - \psi_2)$. These calculations use the same table values as the first-law calculations.

KEY POINT Checklist for any steam problem: (1) fix each state with two independent properties, deciding wet/dry/superheated by comparing with the saturation values; (2) closed system $\rightarrow$ use u and $Q = \Delta U + W$; open system $\rightarrow$ use h and the SFEE; (3) ideal process $\rightarrow s = \text{constant}$; real process $\rightarrow$ isentropic efficiency, then the actual end state from h ; (4) test the result with the second law – $s_{\text{gen}} \geq 0$.

9. Practice problems (answers given)

No.	Problem	Answer
P1	Find the enthalpy, specific volume and entropy of steam at 0.8 MPa with a dryness fraction of 0.9.	$h = 2563.6$ kJ/kg; $v = 0.2164$ m ³ /kg; $s = 6.200$ kJ/kg K
P2	Determine h , v , s and u of steam at 2 MPa and 300 °C, and its degree of superheat ($T_{\text{sat}} = 212.4$ °C).	$h = 3024.3$ kJ/kg; $v = 0.12550$ m ³ /kg; $s = 6.7685$ kJ/kg K; $u = 2773.3$ kJ/kg; 87.6 K
P3	The specific volume of steam at 0.5 MPa is measured as 0.30 m ³ /kg. Find the dryness fraction ($v_g = 0.37480$ m ³ /kg).	$x = 0.80$
P4	Steam at 0.8 MPa is throttled to 0.1 MPa and 110 °C in a throttling calorimeter ($h_2 = 2696.3$ kJ/kg). Find the dryness fraction.	$x = 0.965$
P5	Two kilograms of wet steam, 0.8 dry, at 0.4 MPa are heated at constant pressure until dry saturated. Find the heat supplied ($h_f = 604.7$, $h_g = 2738.1$ kJ/kg).	$Q = 853.3$ kJ
P6	Steam at 1 MPa, 0.95 dry, expands isentropically to 0.1 MPa. Find the final dryness fraction and the enthalpy drop.	$x_2 = 0.836$; $\Delta h = 372.8$ kJ/kg
P7	A rigid vessel of 0.5 m ³ holds dry saturated steam at 0.3 MPa; it is cooled to 0.1 MPa. Find the mass, the final dryness fraction and the heat rejected.	$m = 0.825$ kg; $x_2 = 0.357$; $Q = -1139$ kJ
P8	Steam at 3 MPa and 400 °C expands isentropically in a nozzle to 0.1 MPa. Find the exit dryness fraction and velocity (inlet velocity negligible).	$x_2 = 0.928$; $V_2 = 1199$ m/s

10. Summary and formula sheet

- A **pure substance** has a homogeneous, invariable chemical composition in all its phases. Water changes phase by melting/freezing, vaporisation/condensation and sublimation. **Saturation temperature** and **saturation pressure** are uniquely related; T_{sat} rises with pressure. **Triple point** of water: 0.01 °C, 0.6117 kPa; **critical point**: 22.064 MPa, 373.95 °C, where $h_{\text{fg}} = v_{\text{fg}} = s_{\text{fg}} = 0$.
- Heating at constant pressure: compressed (subcooled) liquid $\rightarrow$ **saturated liquid** (f) $\rightarrow$ **wet steam** $\rightarrow$ **dry saturated vapour** (g) $\rightarrow$ **superheated vapour**. Sensible heat changes temperature; latent heat

(h_{sf} , h_{fg}) changes phase at constant temperature. **Degree of superheat** = $T - T_{sat}$; **degree of subcooling** = $T_{sat} - T$. Evaporation is a surface phenomenon at any temperature; boiling occurs throughout the liquid at T_{sat} .

- **Diagrams:** p - v - T surface (solid, liquid, vapour, three two-phase ruled surfaces, triple line, critical point; the fusion surface leans the other way for water, which expands on freezing); its projections p - T (three curves meeting at the triple point; fusion curve of negative slope for water), p - v and T - v (saturation dome), T - s (heat = area; isentropic vertical) and h - s **Mollier chart** (isobars, isotherms, x -lines, v -lines; isentropic vertical, throttling horizontal).
- **Wet steam:** $x = m_g/(m_f + m_g)$; $v = v_f + x v_{fg}$; $h = h_f + x h_{fg}$; $s = s_f + x s_{fg}$; $u = u_f + x u_{fg} = h - pv$; valid only for $0 \leq x \leq 1$ at saturation. $h_{fg} = T_{sat}s_{fg}$. Superheated steam: (p , T) from the superheat table, $u = h - pv$; compressed liquid $\approx$ saturated liquid at the same T , $h \approx h_f + v_f(p - p_{sat})$. Steam deviates from $pv = RT$ near saturation.
- **Calorimeters:** separating $x = m_g/(m_f + m_g)$; throttling ($h = \text{const}$, throttled steam must be superheated) $x_1 = (h_2 - h_{f1})/h_{fg1}$; combined $x = x_2 m_2/(m_1 + m_2)$; barrel (energy balance with cold water).
- **Non-flow processes:** constant volume $W = 0$, $Q = m\Delta u$; constant pressure $W = p\Delta V$, $Q = m\Delta h$; isothermal $Q = mT\Delta s$; hyperbolic $W = p_1 V_1 \ln(V_2/V_1)$; isentropic $W = -m\Delta u$; polytropic $W = (p_1 V_1 - p_2 V_2)/(n - 1)$; always $Q = \Delta U + W$.
- **Flow processes:** turbine $w = h_1 - h_2$, $\eta_T = (h_1 - h_2)/(h_1 - h_{2s})$; nozzle $V_2 = \sqrt{2(h_1 - h_2)}$, $A = \dot{m}v/V$; throttling $h_1 = h_2$; mixing $h_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2)/(\dot{m}_1 + \dot{m}_2)$.
- **Second law:** s from the tables; $s_{gen} = s_2 - s_1$ (adiabatic) or $\dot{S}_{gen} = \sum \dot{m}s_{out} - \sum \dot{m}s_{in} \geq 0$ decides feasibility; $I = T_0 s_{gen}$; actual turbine exhaust is drier than the isentropic exhaust.

11. Review questions

11.1 Short-answer (two-mark) questions

1. What is a pure substance? Give two examples.
2. What is meant by sublimation? Under what condition does it occur?
3. Give the values of temperature and pressure at the triple point and at the critical point of water, and state what each point signifies.
4. Define the dryness fraction and the wetness fraction of steam.
5. Distinguish wet steam, dry saturated steam and superheated steam.
6. What is meant by the degree of superheat and the degree of subcooling?
7. Define the latent heat of fusion and the latent heat of vaporisation.
8. On what principle does the throttling calorimeter work?
9. How does evaporation differ from boiling?
10. What is the difference between sensible heat and latent heat of vaporisation?
11. Write the expression for the enthalpy of wet steam of dryness fraction x .
12. Write the expression for the specific volume of wet steam. When can v_f be neglected?
13. Write the expression for the entropy of wet steam.
14. Define the internal energy of steam and give its expression for wet steam.
15. Name the instruments used to determine the dryness fraction of steam.
16. Find the enthalpy of wet steam at 0.6 MPa with a dryness fraction of 0.85 ($h_f = 670.5$, $h_{fg} = 2085.6$ kJ/kg).
17. Steam at 0.8 MPa has a specific volume of $0.2 \text{ m}^3/\text{kg}$; find its dryness fraction ($v_g = 0.2403 \text{ m}^3/\text{kg}$).
18. What is the Mollier chart? For what is it used?
19. What are the coordinates of the Mollier chart and which families of lines are drawn on it?
20. What is the significance of the saturation temperature and the saturation pressure?
21. Which process is represented by a vertical line on the Mollier chart? And by a horizontal line?
22. How does the saturation temperature of water vary with pressure?
23. Distinguish between a saturated liquid and a subcooled liquid.

24. Write an expression for the enthalpy of superheated steam in terms of the degree of superheat.
25. Sketch, on a temperature–heat diagram, the conversion of ice at $-10\text{ }^{\circ}\text{C}$ into superheated steam at atmospheric pressure and name the five stages.
26. Why does the throttling calorimeter alone fail for very wet steam?
27. Why is the exhaust of an actual turbine drier than that of an isentropic turbine between the same pressures?
28. State the relation between h_{fg} , T_{sat} and s_{fg} and explain its basis.
29. How is the state of steam (wet, dry or superheated) identified from a given pressure and specific volume?
30. Why does the fusion curve of water on the p – T diagram have a negative slope?

11.2 Long-answer (thirteen-mark) questions

1. Draw the p – v – T surface of a substance that expands on freezing and of one that contracts on freezing, show the p – T and p – v projections with the triple line, the critical point and the saturation lines, explain the significance of the triple and critical points, and derive the expressions for v , h , s and u of a wet mixture in terms of the dryness fraction, stating when they apply.
2. Explain the formation of steam at constant pressure with the help of the T – v and T – Q diagrams, defining sensible heat, latent heat, saturated liquid, wet steam, dry saturated steam, superheated steam, degree of superheat and degree of subcooling.
3. Describe, with sketches, the separating, throttling and combined separating-and-throttling calorimeters, derive the expression for the dryness fraction in each case and state their limitations.
4. Using steam tables, find at 1.5 MPa the saturation temperature, the change in specific volume and in entropy on evaporation and the latent heat; repeat at 0.15 MPa and 8 MPa; verify that $h_{fg} = T_{sat}s_{fg}$ in each case and explain the trend with pressure.
5. Steam at 1.5 MPa has a specific volume of $0.10\text{ m}^3/\text{kg}$. Identify the state and find its dryness fraction, enthalpy, entropy and internal energy; if it is heated at constant pressure until $v = 0.14\text{ m}^3/\text{kg}$, find the final temperature and the heat supplied.
6. Find h , s , v and u of steam at 1.2 MPa and $350\text{ }^{\circ}\text{C}$ from the tables and state the degree of superheat; compare v with the ideal-gas value ($R = 0.4615\text{ kJ/kg K}$) and explain the deviation.
7. A vessel of 0.05 m^3 holds a mixture of saturated water and steam at $240\text{ }^{\circ}\text{C}$ with 8 kg of liquid. Find the pressure, the total mass and dryness fraction, and the specific volume, enthalpy, entropy and internal energy of the mixture.
8. Eight kilograms of water at $40\text{ }^{\circ}\text{C}$ are heated at 1.2 MPa to $320\text{ }^{\circ}\text{C}$. Show the process on T – s and p – v diagrams, find the changes in volume, enthalpy, internal energy and entropy, the heat supplied and the work done, and verify the first law.
9. A rigid vessel of 0.7 m^3 contains 1 kg of steam at 2.5 bar. Find v , T , x , u , h and s ; the vessel is cooled to 1 bar – show the process on the T – s diagram and find the final dryness fraction and the heat rejected.
10. Steam at 0.35 MPa and $250\text{ }^{\circ}\text{C}$ is cooled at constant volume. Find the temperature at which it becomes saturated, the dryness fraction at $75\text{ }^{\circ}\text{C}$ and the heat transferred per kg.
11. Steam at 0.6 MPa and $300\text{ }^{\circ}\text{C}$ in a rigid vessel is cooled to $50\text{ }^{\circ}\text{C}$. Show the process on T – v and p – v diagrams, find the state at which condensation begins, the final dryness fraction and the heat rejected, and compare the heat removed before and after condensation begins.
12. Steam at 1.4 MPa and $300\text{ }^{\circ}\text{C}$ expands isentropically in a turbine to $45\text{ }^{\circ}\text{C}$. Sketch the process on T – s and h – s diagrams and find the exhaust pressure, the exit dryness fraction and the ideal work.
13. Steam at 2.5 MPa and $350\text{ }^{\circ}\text{C}$ expands in a turbine to 0.06 MPa. Find the exit dryness fraction and the work for isentropic expansion; for an isentropic efficiency of 82 % find the actual work, the actual exit dryness and the entropy generated, and explain why the actual exhaust is drier.
14. Superheated steam at 0.9 MPa, $250\text{ }^{\circ}\text{C}$ at 1 kg/s mixes adiabatically with wet steam at 0.9 MPa, 0.93 dry, the mixed flow being 2.5 kg/s; find the state after mixing. The mixture expands isentropically in a nozzle to 0.45 MPa; find the exit dryness fraction and velocity.

15. Superheated steam at 1.2 MPa, 300 °C (2 kg/s) mixes adiabatically with wet steam at 1.2 MPa, 0.92 dry (1.2 kg/s). Find the enthalpy and temperature after mixing and the rate of entropy generation, and state with reasons whether the process is possible.
16. Steam at 1.2 MPa and 250 °C expands isentropically through a nozzle to 0.35 MPa. Neglecting the inlet velocity find the exit dryness fraction, the exit velocity and the exit area for 0.6 kg/s.
17. Explain how the first and second laws are applied to non-flow and flow processes of steam, and show with the help of the T–s and h–s diagrams how isentropic efficiency and entropy generation are evaluated for a turbine, a nozzle, a throttle valve and a mixing chamber.

Text: P. K. Nag, "Engineering Thermodynamics", 6th Edn., McGraw-Hill — Chapter 9. References: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.; R. K. Rajput, "A Text Book of Engineering Thermodynamics", 5th Edn. Steam properties: IAPWS-97.